

FACULTY OF
COMPUTER SCIENCE

Otto-von-Guericke-University Magdeburg

Faculty of Computer Science

Department of Databases and Software Engineering

Comparative Analysis of Multimodal Emotion Detection Approaches

Bachelor Thesis

Author:

Lukas Eichel

Examiner:

Prof. Dr. rer. nat. habil. Gunter Saake

2nd Examiner and Supervisor:

Dr.-Ing. David Broneske

Magdeburg, 13.09.2024

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Contents

1. Introduction **1**
- 1.1. Motivation . . . 1
- 1.2. Structure . . . 2

2. Background **3**
- 2.1. Emotion . . . 3
  - 2.1.1. Emotion Classification . . . 4
  - 2.1.2. Emotion in Text . . . 5
  - 2.1.3. Emotion in Speech . . . 6
  - 2.1.4. Emotion in Facial Expression . . . 7
- 2.2. Classification Approaches . . . 8
  - 2.2.1. Classification in Text . . . 9
  - 2.2.2. Classification in Audio . . . 14
  - 2.2.3. Classification in Images and Videos . . . 17
- 2.3. Datasets . . . 20

3. Related Work **23**
- 3.1. Audio and Video Based . . . 23
- 3.2. Audio, Video and Text Based . . . 24

4. Design and Implementation **27**
- 4.1. Unimodal Text Classification . . . 27
  - 4.1.1. Preprocessing and Prompt Engineering for LLM . . . 29
- 4.2. Unimodal Audio Classification . . . 32
- 4.3. Unimodal Video Classification . . . 36
- 4.4. Multimodal Classification . . . 40

5. Evaluation **43**
- 5.1. Setup . . . 43
  - 5.1.1. Accuracy Metrics . . . 44
- 5.2. Results . . . 44

6. Conclusion **47**
- 6.1. Summary . . . 47
- 6.2. Future Work . . . 47

Bibliography **49**

Chapter 1.

Introduction

1.1. Motivation

Artificial intelligence and respectively deep learning architectures are currently beeing developed quite fast this time and optimized more and more. With the introduction of transformer models user-oriented applications to generate text, audio and videos or identify everdays sequences featuring trainable logic. Furthermore they are declared as all-rounder tools, capeable to slowly making the first jobs obsolete which shows the question if they are really able to work as a multi-tool or if they just take profit from the large range of training data and if there will be some kind of bias due to coarse data with less depth and no focus on a special task like emotion detection.

From an early stage psychological approaches are showing fixed behavioral patterns when showing emotions and sentiments including face and voice actions regarding semantic and syntax. Basic math and machine learning algorithms were able to classify some information and identify emotion thus neural networks are bringing more performance, better results in less time. The introduced transformers are infamous for their contextual understanding as well as very good and consistent outcomes.

This leads to the question if already developed applications like ChatGPT or in general approaches based on deep learning are able to recognize emotion from video, audio and text in the context of static interviews and real-life excerpts. The goal of this thesis is to compare state of the art models and architectures mostly focused on transformers which are present in every modal. Further multimodal emotion detection might be a topic to analyize different feature fusion models and if the differnt labels from the different architectures might lead to an more accurate result than a single modal outcome. This topic leads from basic voting strategies up to further more complex architectures which are using the logits, kernels or other raw output from the models. A comparison can be done to show if the effort is necessary for an accurate result or if given basic methods are nearly as good or even better. These points sequels into an assumption that bimodal emotion classifcation can be more efficient and asymptotically accurate as well as trimodal classification which might show that one of the modals might have a small impact on the results but not an crucial one. Intuitively this can be emotion classification with features from text due to the impact of irony or other vocal features bringing a not unambigious meaning for sentences or words related to their emotion label. The opportunity to ignore one or more modals offers several optimizations as for example one large architecture

less is needed or the choosen feature combination method has a reduction in complexity and computation time.

In general emotion detection brings an interesting field of opportunity for research regarding machine-based simulation of human behavior and understanding of interactions related to interviews or conversations. Also after learning from the features most generative models are able to reproduce emotion along the trained information giving perspective in robotics or other forms of real world imitations. Current research still focused on the general, non-transformer based models like convolution, reccurent networks or even simpler machine learning approaches. Due to limitation from copyright or closed-source projects a few architectures are reproduced but aren't trained one the original datasets which might lead to different accuracys in comparison to the usage of the original model with the pretrained weights.

1.2. Structure

This work starts by showing and discussing the fundamental, psychological backgrounds in Chapter 2 for the term emotion and which features are necessary to identify them. Further the classifcation approaches in machine and deep learning are presented showing their unsupervised and supervised methods for training and feature extraction. At the end of this chapter the datasets for the further implementation parts are introduced. Related works in Chapter 3 are presented to display current research on multi-modal detection and their different methods and learning proceedings. Continuing with the design and implementation of the different supervised models, Chapter 4 presents self-trained and also pretrained models and how to expand as well as finetune their knowledge. In addition the architectures to fuse the information for multi-model emotion detection are shown and different fusion methods are discussed including some kinds of basic scoring and more advanced feature combinations. These accomplishments are evaluated in Chapter 5 introducting measurement criteria for relieable and comparable results. At the end this thesis is summarized and gives an overwiev in comparison to some outcomes from related work concluding with a look on possible future work in emotion detection.

Chapter 2.

Background

This chapter will give an introduction and overview on different approaches to classify types of emotions with respect to their advantages and disadvantages. Furthermore it will explain the approaches and assumptions to recognize emotion in text, video and audio. Based on this common implementation techniques for supervised and unsupervised approaches are presented with a focus on convolution and transformer models since they are the leading architectures in deep learning. At the end, datasets are introduced which are the foundation for the later implementation and evaluation of the models.

2.1. Emotion

There is no unambigious definition for the term ‘emotion’. There are many different emotions, with different level of intensity. Considering the question of what an emotion is and where it comes from, William James discussed two different theories about the origin of them. Either there are seperate areas in the cerebrum or else they correspond to motory and sensory processes [1]. If they originate from the seperate areas, the research would be much easier. They could observate these regions while testing different actions and influences on the probants. The results should be quite similar between most people. However it is most likely the case, that emotion correspond to these motory and sensory processes. They will be different from human to human in intensity and execution, so researchers have to study the small similarieties and also the possible patterns and variations from the shown processes. In this case these factors and all brain activities must be captured to set a prove for it. Many work in psychology add further aspects to the theory. In order to provide a framework, the following chapters will focus just on the given emotion and how to recognize it in written text, speech and facial expressions. These are the most common expressive features captured by emotion theorists, but there are many more like gestures or going into a medical direction the blood pressure and other factors might have an approach for determining the leading emotion.

In the other direction there are first approaches to simulate sentiments with computer or robots like animatronics. Learned patterns can be applied from a simple command to emotion features which are currently still very overacted.

Figure 2.1 presents that information like motory, sensory processes, their combination and the differentiation from semantic and syntatic context are categorized in high-level and low level information to process an emotion.

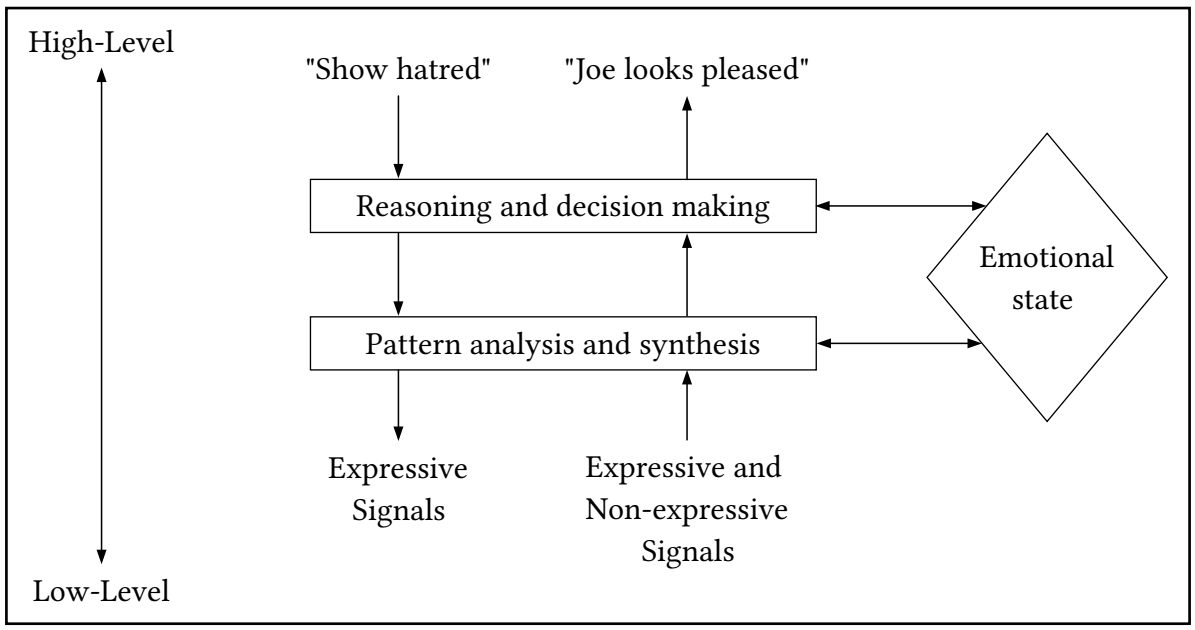


Figure 2.1.: Information flows from both high to low to high in a system that can express, recognize and "have" emotions [2].

High-level information are defined by the semantic of a situation which can be drawn from audiovisual processes for example the command "Show hatred". To show hate a certain emotional state has to be reached and must be reasoned by the situation. As a conclusion low-level information, for instance face expressions, gestures or other syntactic articulations can be observed from the person or system wich executes the given instruction. These patterns are analyzed or learned and synthesized by the process before.

On the other direction, these low-level expressive information can be combined to get the semantic of the emotional state. In addition there can be non-expressive data as a valid input. It describes the passive influence on a persons state like the weather or the environmental conditions. Most datasets won't cover this type of information.

The mid-level builds patterns for the given input and combines different low-level signals to synthesize better reasoning for a high-level output. For example a slightly faster speech rate and a tense articulation can be clustered together to get classified as an pleased look.

2.1.1. Emotion Classification

There are several apporaches to classify the semantic high-level features. The most popular classification is the basic-emotion framework by Paul Ekman [3] describing six different emotions: anger, disgust, fear, surprise, sadness and happiness. This arose from various facial expression, which Ekman categorized to the corresponding emotion.

Over the time diverse frameworks were built on this foundation to introduce for each basic emotion multiple detailled emotions. A common model is the Plutchik wheel of emotion, which is conventional used to classify emotion in text. It offers eight types of basic emotion: joy, trust, fear, surprise and their opposites sadness, disgust, anger and anticipation [4].

Visible in Figure 2.2 each emotion has multiple levels of intensity. For example, the enhanced level for disgust is loathing, a more moderate level is boredom. In total there are 24 different emotions. A variety of sentiments and their intensitys lead to a proper illustration of the real value.

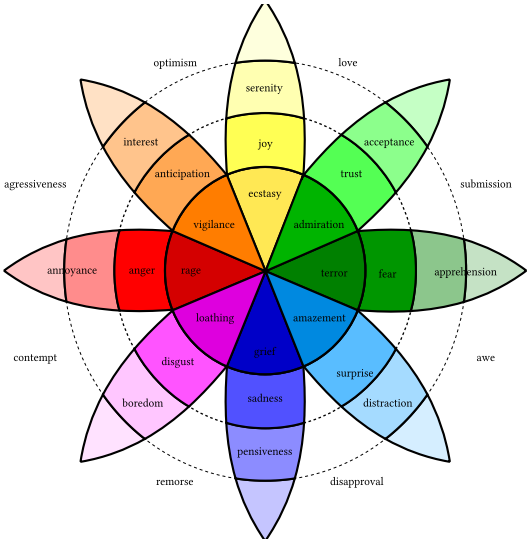


Figure 2.2.: Plutchik Wheel of Emotion [5].

Although there will be a loss of detail, the content of this thesis will focus on the basic emotion from Paul Ekman. The wheel of emotion from Plutchik is suitable for emotion detection in text but most datasets in facial emotion recognition are focused on Ekmans emotion theory. As shown later in this thesis, many multimodal datasets are based on the work from Ekman. It is harder to distinguish all emotion from Plutchik in audio processing and face recognition.

2.1.2. Emotion in Text

The most common approach for classifing emotion from natural language text is to classify the semantic content of these sentences. In order to make this possible, two syntactical assumptions are needed.

1. There is a function that is able to map words and sentences to their corresponding emotion.
2. Words and sentences do not change their meaning over time.

The first assumption can be approached by training a deep neural network with a suitable dataset or a clustering algorithm that measures the similarity between words and sentences to match their corresponding emotion label. The function must be surjective, so a token won't be able to be labeled with more than one emotion at the time. The second assumptions is important for a relieable result in training but it is also the flaw from emotion detection in text. For example the sentence "I'm glad to be here". Expressions like irony, emphasis in the voice or gestures can change the proper meaning of the sentence. It may can be interpreted as happiness or as disgust. To take this into consideration further observations of other expressive and non-expressive signals are needed.

2.1.3. Emotion in Speech

Emotion in audio is in contrast to the classification in text mainly focused on the syntax from the recording. The actual signal input is represented as a waveform and sinoid functions. A single wave is defined by

$$g(t) := A \sin(2\pi(\omega t - \varphi)) \tag{1}$$

A complete waveform as presented in Figure 2.3 and Figure 2.4 consists of many different sinoid functions which are added and concatinated together [6].

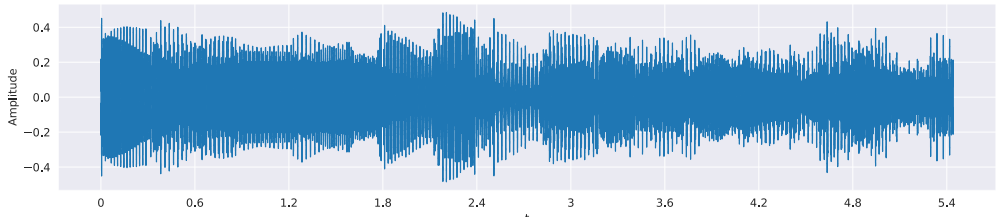


Figure 2.3.: Waveform of the first seconds of an audio recording

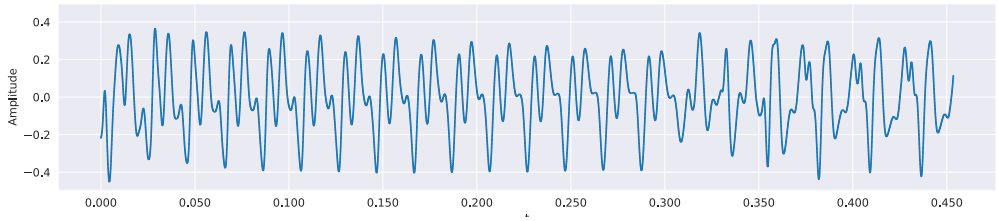


Figure 2.4.: Enlargement of the first half second from Figure 2.3

Frequency (ω) measures the pitch from the audio and especially from the voice. As shown in Table 2.1 multiple features can be extracted from the frequency respectively the pitch. From a measured neutral foundation changes like an average pitch, the range or other changes can be observed. For example a person which speaks in a very much higher pitch, with a wide variablity in the range and changes the frequency abrupt on stressed syllables, then this person feels most likely anger.

Amplitude (A) shows the intensity and the loudness of the given audio. Furthermore it is also an important feature and easy to extract from speech with the help of an existing neutral foundation.

Phase (φ) shifts $g(t)$ along the x-dimension. This parameter is not important in the feature measurement but has a significant role in audio decomposition which is described later.

Extracting features like speech rate, articulation or voice quality requires more observation. Speech rate can be measured with words per second, sentences per minute or other similar metrics. Normally when looking at waveforms the intermissions betweens sentences or words can get retrieved. A higher rate is equivalent to more observed gaps in the same time of a normal speech speed. Articulation and voice quaility is hard to track without a semantic knowl-

edge. Therefore a combination of the already presented features is needed which can be done by pattern matching and comparison to the neutral foundation.

Emotion / Feature	Fear	Anger	Sadness	Happiness	Disgust
Speech Rate	much faster	slightly faster	slightly slower	faster or slower	very much slower
Pitch Average	very much higher	very much higher	slightly lower	much higher	very much lower
Pitch Range	much wider	much wider	slightly narrower	much wider	slightly wider
Intensity	normal	higher	lower	higher	lower
Voice Quality	irregular voicing	breathy chest tone	resonant	breathy blaring	grubled chest tone
Pitch Changes	normal	abrupt on stressed syllables	downward inflections	smooth upward inflections	wide downward terminal inflections
Articulation	precise	tense	slurring	normal	normal

Table 2.1.: Features of an audio with correlating emotion [2], [7].

2.1.4. Emotion in Facial Expression

Besides his theory about the six basic emotions, Paul Ekman introduced a system to map different face expressions to their relating sentiments by combining changes in different areas of the expression field. This facial action coding system (FACS) is based on these movements which are called action units. There are 28 main actions which can be mapped to the six main emotion. In the following of this thesis, they will be used as the main features.



Neutral Sad Fear Disgust

Figure 2.5.: Examples with their annoations from the CREMA-dataset [8] which show emotions based on facial expression features.

Visible in Figure 2.5 are distinctive combinations of face features are able to represent different emotion. For example to observe disgust, commonly the typical wrinkling of the nose, the contraction of the eyebrows and a lifting of the upper lip can be watched. Additionaly to the movement of the eyebrows, sadness is marked through the downturned corners of the mouth. Thus a face and his expressions can be broken down into seperate regions.

Eyes (upper face): The most action units to recognize emotion within facial expressions are defined by movement near the eyes. Eyebrows are seperated into an inner and outer part for a more detailed classifcation. Furthermore they can be raised or pulled together leading into four different states. Besides from eyebrows, eyelids are another meaningful part in that area. They can be tensed, raised by opening the eyes wide and lowered by narrowing the eyes. In total this region plays an important role for every basic emotion by Ekman and a relevant part for the most sentiments from Plutchiks theory.

Nose: This face region only impacts one basic emotion, disgust, shown in Figure 2.5. This is due to the limited range of movement of this part of the head. Thus this section of the face won't differ from the upper part of the face and the lower part described above due to its low occurrence.

Mouth (lower face): The cheeks and the chin are also correlated to this area for the following work in this thesis due to their tight connection to the lips and mouth by means of muscles. This leads to a total of 16 different action units due to the numerous possibilities, e.g. deforming the lips, opening and closing the mouth and the resulting movements in this area.

In his research Ekman defined each action unit and their corresponding emotion in detail which can be looked up in his book about the basic FACS System [9].

Furthermore in 2002 [10] he updated his idea of the facial expression system and integrated more action units adding up to more than 40 possible features in total. This includes tilting and rotating the head and more details on the upper face like winking or blinking with the eyes.

In addition a closer look at the various intensities of an action are presented and classified into 5 levels: trace (A), slight (B), marked or pronounced (C), severe or extreme (D) and maximum (E). Most common are slight or marked and pronounced, while the higher intensities are visible in films or other acted illustrations. Maximum intenseness is the most rare but can be found in theater or generally live action plays. The different letters are used as an indicator together with the numerated action unit for a better description and overview of all possible combinations.

2.2. Classification Approaches

Over time many approaches to detect and classify emotion have been developed. There are severeral supervised architectures and unsupervised techniques which represent the current state of the art for emotion detection in text, audio and video.

2.2.1. Classification in Text

To classify emotions in text, multiple preprocessing steps are necessary due to the lack of numerical values. In the following some ideas for preprocessing are presented. Furthermore, a look at the various classification methods will be taken. Here a special focus on the supervised methods is applied due to better comparability of already existing and pretrained deep learning models.

Word Embedding and Tokenization

The most important part of all classification approaches is called word embedding. It represents words as real-valued vectors [11]. It should use the syntactic but also the semantic meaning of the words to get a useful vector. This technique is commonly used in natural language processing like information retrieval in text.

A naive approach to generate such an embedding function would be a one-hot vector encoding [12]. It represents every word in a space as an different vector. For example the sentence:

The tree is green.

will be first tokenized into a set of words V.

$$V = \{\textit{the}, \textit{tree}, \textit{is}, \textit{green}\}$$

The size of the set $|V| = 4$. As a result a mapping $w \in V$ to a $|V|$-dimensional vector is needed.

$$A : V \to \mathbb{R}^{|V|}$$

$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$
the	$\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$	tree	$\begin{pmatrix}0\\1\\0\\0\end{pmatrix}$	is	$\begin{pmatrix}0\\0\\1\\0\end{pmatrix}$	green	$\begin{pmatrix}0\\0\\0\\1\end{pmatrix}$

Table 2.2.: Possible one-hot vector mapping for the sentence “The tree is green.”

To measure the similarity the cosine similarity is often used [12], [13]. Similar words or synonyms should have a smaller angle between each other, than unsimilar words.

$$\text{sim}(w_1, w_2) = \cos(\theta) = \frac{w_1 \cdot w_2}{\|w_1\| \, \|w_2\|} \tag{2}$$

Because the one-hot vector embeds it word by word without any context, there won’t be any similarity or dissimilarity between the words. This is also the reason why simple one-hot encoding is less used in practise. A semantic consideration will be more useful.

The Continous Bag-of-Words Model, shortly CBOW is capable to take the context of a word as an input to calculate an embedded vector with emphasis on syntax and semantic. It describes a neural network which mainly consists of three layers: the input layer, a hidden layer and the output layer [14].

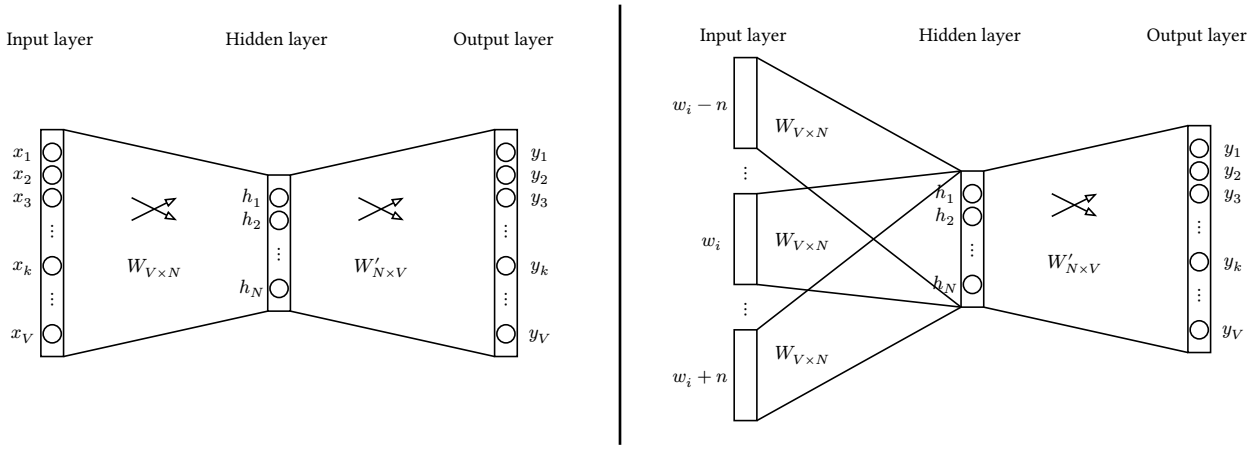


Figure 2.6.: Left: Simple CBOW-architecture for a single word; Right: CBOW-architecture for multiple words. Words have length k and are padded to size V to match the dimension [12].

The architecture shown on the right side of Figure 2.6 takes one one-hot vector $A(w_i)$ as an input expression. To get the context of the inputted term, phrases before and after are needed. Therefore an addition to the architecture is necessary.

With unsupervised learning the model on the right of Figure 6 is able to generate a new vector which represents the context of a word. Now, under consideration of the cosine similarity, words with a similar context have a quite smaller angle compared to without any connection in semantics. In both architectures, the hidden layer acts like an arithmetic operator which combines the context to the outputted one-hot vector.

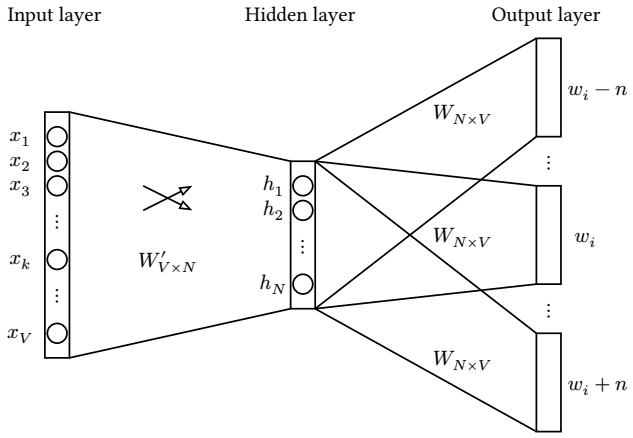


Figure 2.7.: Skip-Gram Model. All words have size V (after padding) [15].

The Skip-Gram Model calculates an embedding in reverse from the idea of the CBOW-Model. As the architecture shows in Figure 2.7 it tries to calculate the surrounding words with a given target term [16]. Therefore a training like the CBOW with contextual phrases is needed before it can work like a semantical decoder. Both architectures are commonly used in different pretrained word embedding networks, like Word2Vec from Google [15], [17].

Fast WordPiece Tokenization is an approach to tokenize sentences and words in an efficient way by splitting words into smaller subwords. It is used in a language understanding transformer model from Google which is called Bert [18].

The basic problem from other tokenization methods like the normal WordPiece Tokenization starts in the vocabulary tokenization in tries. Trie memory is a paradigma which is able to store strings efficiently where each edge of the data structure stores phonemes. Traversing a path consisting of multiple edges results in a word. As a result the structure does not have to store redundant parts of words [19].

The tokenization has to search for the longest token at a position. Therefore it starts from the shortest substring and iterates over the input text, while trie matching to match the longest prefix which matches a vocabulary token [20]. If the trie matching fails at the next character, the algorithm has to involve backtracking because the longest vocabulary token may be several characters back. This results into a time complexity of $O(nm)$ where m is the maximum vocabulary token length and n the input length.

Fast WordPiece Tokenization introduces so called failure links and pops. Failure pops are a list of longest-matching prefix tokens of a node in the trie, which can be removed in order until a failure link node is reached. The failure node is able to represent the missing suffix. This whole algorithm called LinMaxMatch has a time complexity of only $O(n)$.

Supervised Classification Approaches

With a given word embedding and tokenization supervised learning approaches are able to recognize emotion in text. Therefore an annotated dataset is needed, which has a relieable mapping from sentence or word to the corresponding emotion.

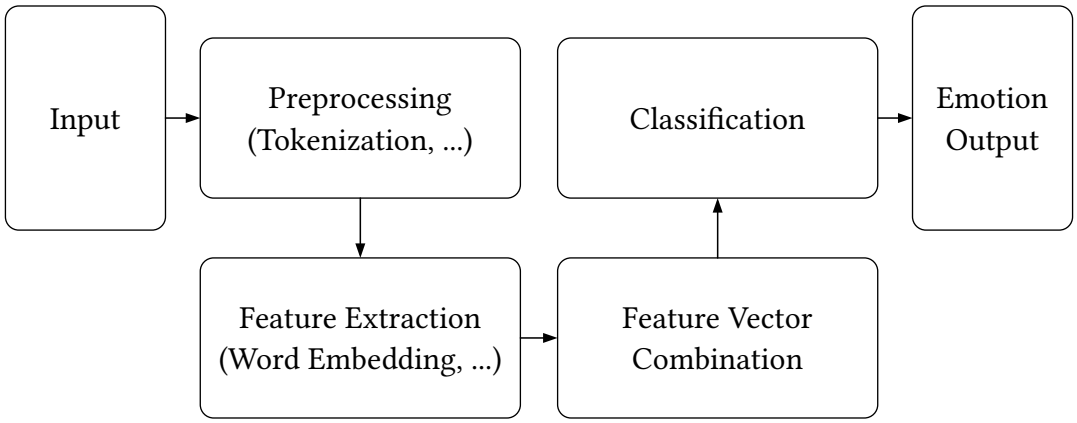


Figure 2.8.: Deep Learning Assisted Semantic Text Analysis

Most models are using the architecture paradigma shown in Figure 2.8 . They mostly differ in the step of combining the feature vectors and the type of classification because it offers the biggest room to maneuver. A preprocessing and feature extraction is always necessary. Sometimes a WordNet [21] is integrated to get a better overview about synonyms or to paraphrase words.

Convolutional and Recurrent Neural Networks (CNN/RNN) are an effective approach on sentiment analysis of text [22]. Convolution consists of a kernel-function which downscales the input but keeps the important features. This technique is called Pooling and brings several benefits like less memory usage, less overfitting in training and a faster computation time. Herefore multiple words or sentences of the space V are combined to region matrices. On each region L convolutional filters are applied to learn local n-gram features.

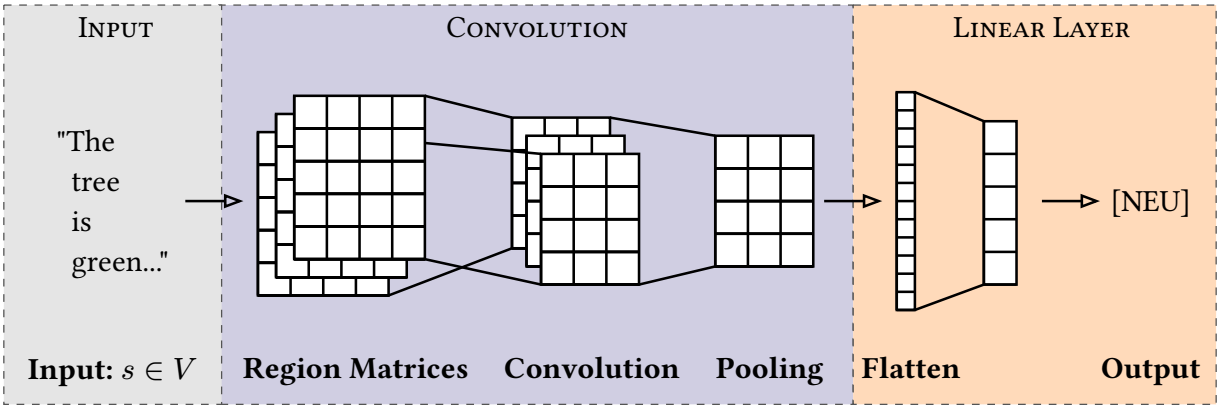


Figure 2.9.: Simple Convolution based Neural Network to predict an emotion from a given subset of sentences

Figure 2.9 shows such network with the convolutional part. In common architectures multiple convolution and pooling layers are stacked together with an activation function like ReLu in between of each block which improves the output by setting constraints to the convolution outputs. The constraint which is effected by ReLu is the prohibition of negative values by the function $f(x) = \max(0, x)$. This subsequency of the convolution blocks leads to a hierarchical decomposition of the given embedded region matrix. As a result multiple features can be found from each block and each convolution kernel which might result in a higher accuracy. At the end an ordinary decoder in form of a simple linear layer with an activation function like SoftMax is utilized to calculate the resulting emotion from the pooled and flattened vector. Normalization like layer norm or batch norm between each layer can help to ensure better propagation and reduce numerical instability [23].

Further optimizations to fundamental neural network architectures are gating mechanisms from recurrent networks and the bi-directional flow in the model forwarding architectures. They introduce new types of storage cells which allows to store long term features [24]. In the context of emotion detection in text, it might store important context from previous sentences and words to differ between fill-words like “this”, “with” and emotionally loaded words. Commonly used are long short term memory cells (LSTM) [25] and gated recurrent units (GRU) [26] which both try to prevent gradient vanishing which is a big problem in recurrent networks. It occurs while propagating backwards through time. Due to many multiplication steps, the gradient converges to 0. These two architectures differ mostly in their structure than in their functionality. Additionally to the GRU cell, an LSTM has two more control gates which gives more possibilities to constrain and rule the memory inside the structure. In both cases they can improve learning features from context and time series.

Transormers are a deep-learning architecture which is based on multi-head attention.

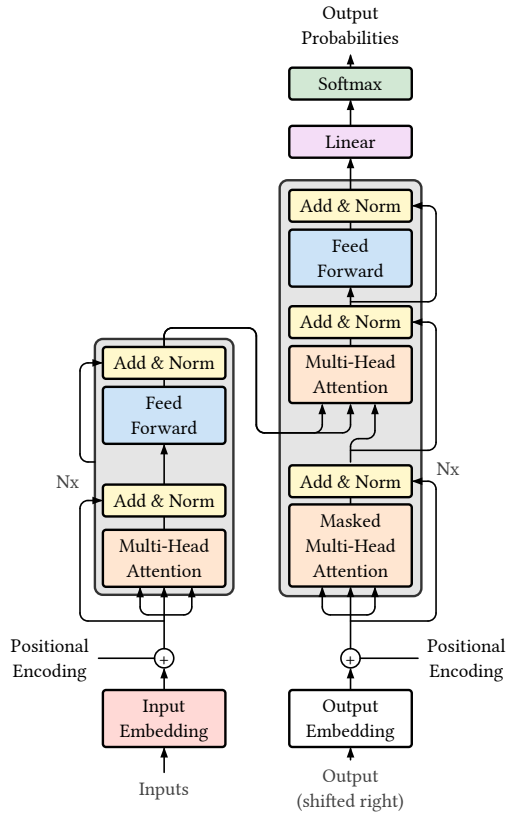


Figure 2.10.: Common transformer model architecture with multi-head attention [27].

The basic idea of attention solves a huge bottleneck in common neural network architectures because fixed input vector sizes may result in a large overhead. In the example of natural language processing, this could be a result from very large sentences. Attention mechanisms try to solve this issue by focusing the input on to specific parts e.g. training the input weights on important words or subsets from the given sentence.

The resulting transformers are currently setting a milestone in natural language processing and providing the basis for ChatGPT from OpenAI [28] or other text understanding models, because of their scalability due to the attention mechanisms and their reliable results. Figure 2.10 shows such a transformer architecture. An input sentence will be fed into the model where it first gets positional encoded thus the position and the frequency for every word in the sentence is stored.

Now, the input vectors are passing through the first multi-head attention layer to lay the focus on important parts of the input. Neverless, the actual input and the weighted part are added and normalized together which results in the original sentence but with tagged highlights due to the attention. To learn the features for the output token, the input traverses through a feed forward network which consists of basic uni-directional layers. Likewise the input and the output are matched together with a residual connection. The same proceding is repeated but with an autoregressive addition of already calculated output. But in contrast to the regular multi-head attention a masking in this process is introduced which sets the weight for unknown or respectively future tokens to negative values thus the model can learn realistic predictions only with already known tokens. The output of the transformer architecture are probabilities for the next state. The already mentioned autoregressive implementation gives the possibilities to predict a long sequence of tokens.

In addition to the approaches with neural network architectures like transformers, other supervised methods like support vector machines, decision trees or naive bayes are used. These are not covered in this thesis as it exceeds the scope of this work and are a possible part for future work.

In summary, supervised methods follow the basic principle to learn features along a given target. Therefore enough training data is needed with the additional annotated labels.

At the end even the differentiation between convolution, recurrent or transformer networks will not make any noticeable changes in measurements if the architectures and models are large enough [29]. In literature this "bitter lesson" [30] is researched step by step and also shown for language models, convolution nets and transformers.

"Scaling Laws for Neural Network" [31] shows the improvement of the loss, when a model has many parameters, much computation time in training and a large dataset. Therefor this thesis will consider the transformer models as a focal point for later evaluation, because there are many large pretrained language models available which are able to recognize emotion and are trained and developed to satisfy almost all three factors. Language modeling performance improves smoothly as the model size, datasetset size, and amount of computation used for training is increased. For optimal performance all three factors must be scaled up in tandem.

Unsupervised Classification Approaches

Many datasets do not include annotated targets or subsets of the data cannot be used as a prediction set. In unsupervised learning, the algorithms are trying to extract patterns from this data which are mostly similarities or differences in the training set.

A common unsupervised approach in natural language processing for extracting keywords or similarieties from text is called Latent Semantic Analysis.

Latent Semantic Analysis (LSA) [32] is an information retrieval method in text based on singular value decomposition. The algorithm first transforms every word from a sentence into a matrix M. This matrix can be dismantled into a diagonal matrix S and two orthogonal matrices U, V through

$$M = U \cdot S \cdot V^T \tag{3}$$

S contains singular values which can be easily converted to eigenvalues. These are used to process a query on the vector space to find subsets of M which fullfills the query by applying the cosine similarity. This whole method is quite expensive in computation time because of the necessary decomposition and dimension reductions [33].

There are many more unsupervised approaches for emotion detection and information retrieval in text [34], [35] but they won't be part of the following comparisons and benchmarks as they also exceed the scope of this thesis. Furthermore, no significant improvements and differences are expected in comparison to the presented unsupervised approaches and also the shown supervised classification methods due to the scaling laws for neural networks.

2.2.2. Classification in Audio

Similar to classification in text as shown in Chapter 2.2.1 there are some approaches to supervised and unsupervised methods out there. First of all an encoding for the audio is necessary to ensure the extraction of the necessary features.

Fourier Transformation

To break up the audio signal into multiple atomic sinusodial functions, a fourier transformation and analysis can be applied. For each frequency $\omega \in \mathbb{R}$ from the given signal, the transformation yields a coefficient d_ω and a phase φ_ω from which a sinusodial oscillation can be retrieved. Looking back at the equation from Chapter 2.1.3 a normalization with regard to their amplitude is collected by assuming $A = \sqrt{2}$.

$$g(t) := A \sin(2\pi(\omega t - \varphi)) \tag{4}$$

For each frequency and phase the normalized function $\cos_{\omega,\varphi}(t) : \mathbb{R} \to \mathbb{R}$ is given by:

$$\cos_{\omega,\varphi}(t) = \sqrt{2} \cos(2\pi(\omega t - \varphi)) \tag{5}$$

Due to the idea of approximating the original audio function with sinusodial waves, a first approach to the fourier transformation can be obtained by maximizing the similarity between both signals.

$$\begin{aligned} d_\omega &:= \max_{\varphi \in [0,1)} \left(\int_{t \in \mathbb{R}} f(t) \cos_{\omega,\varphi}(t) \, \mathrm{d}t \right) \\ \varphi_\omega &:= \operatorname{argmax}_{\varphi \in [0,1)} \left(\int_{t \in \mathbb{R}} f(t) \cos_{\omega,\varphi}(t) \, \mathrm{d}t \right) \end{aligned} \tag{6}$$

Bringing this into the context with polar coordinates and the additional idea to store both values d_w and φ_ω into a single complex value. This results in the common fourier transformation form given by:

$$\begin{aligned} \hat{f}(\omega) &= \int_{t \in \mathbb{R}} f(t) \exp(-2\pi i \omega t) \, \mathrm{d}t \\ &= \int_{t \in \mathbb{R}} f(t) \cos(-2\pi\omega t) \, \mathrm{d}t + i \int_{t \in \mathbb{R}} f(t) \sin(-2\pi\omega t) \, \mathrm{d}t \end{aligned} \tag{7}$$

These equations are however only suitable for non-discrete signal-processing and must be adapted to a discrete fourier transformation. A sampling period T is introduced to sample the signal in multiple time steps. An approximation for $\hat{f}(\omega)$ is obtained

$$\hat{f}(\omega) \approx \sum_{n \in \mathbb{Z}} T(f(nT)) \exp(-2\pi i \omega n) \tag{8}$$

with a time complexity of $O(N^2)$ where N denotes the length from the given signal [6]. Thus N can get quite large some approvements are implemented. For example the fast fourier transformation with a complexity of $O(N \log N)$, uses a divide and conquer algorithm to split the signal, calculate the fourier transformation and reassemble it [36]. Also common is the short time fourier transformation which divides the audio into windows with a given size n and a hop size to determine the step size of the sliding window [6], [37].

Supervised Classification Approaches

Analog to Chapter 2.2.1 the supervised classification approaches follow the same structure. Recurrent- and convolution-based networks are the most common approach [38], [39] to classify features from audio after retrieving the fourier representation for every signal and can also be applied to emotion. Continuing from the already presented fourier transformation and analysis, these supervised classification methods refer to the feature extraction from spectograms [40], [41] which is the visual representation of the given signal and a lightweight implementation for these networks.

Meanwhile the first transformer based audio classification are presented [42]. The idea is an amalgamation of the idea from convolution architectures and the common transformer model.

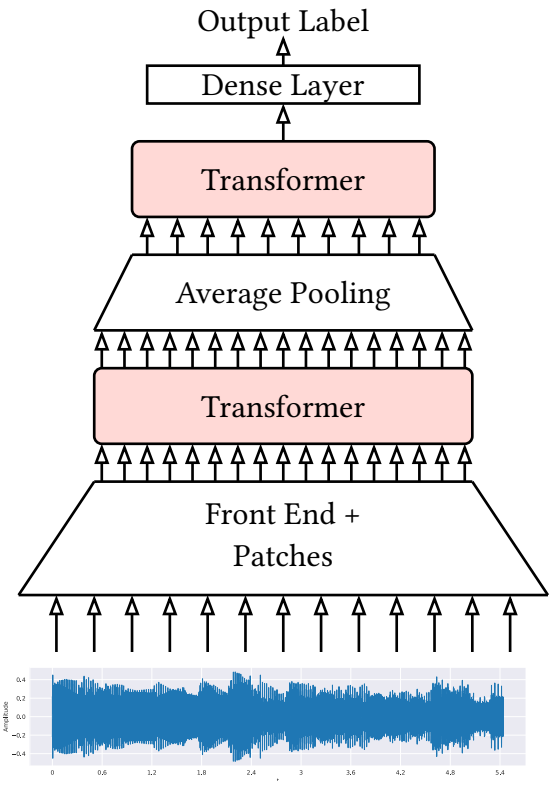


Figure 2.11.: Audio transformer model

Therefore it inputs the raw audio data without any preprocessing besides from padding to the right length. As shown in Figure 2.11 the model first divides the signal into multiple fixed-length patches to feed them into a front-end layer. The patches describe very short sequences of the audio and show the resolution in which the work will subsequently be carried out. It consists of some dense layers which process the raw audio to multiple feature vectors. These are suitable for learning in the common transformer architecture. Furthermore they are chained together with average pooling layer, giving the convolution like downscaling and compressing effect. This step can be repeated multiple times until the required output shape is nearly matched. At the end a decoder layer in form of some dense layers is applied to predict the resulting label. Equivalent to the already introduced convolution networks, activation functions like ReLu or softmax can be interposed between all existing layers for more stability.

Nonetheless the bitter lesson from the previous chapter also applies here. Therefore and because of the already large scope of this thesis, further considerations about other supervised approaches like SVM or decision trees, are not made.

Unsupervised Classification Approaches

Unfortunately unsupervised audio information retrieval approaches are quite rare. There are a few methods which use k-means to learn from local audio features [43]. Thus they are not really suitable for emotion detection due to their low capability of detecting global features. However in spite of that there are some methods to collect local features which can assist the supervised deep learning method [44].

2.2.3. Classification in Images and Videos

Relating to the previous classification methods, the foundation for emotion detection in images and videos is quite similar. However, there is much research to optimize the supervised and unsupervised strategies regarding the input size and computation time. Furthermore video has interesting advantages and disadvantages towards images, which is necessary for the following methods and optimizations. The disadvantages comes with the additional dimension time, which is introduced through videos as a time series while they can be dissected into single images.

Supervised Classification Approaches

As always convolution models are the most common architectures for deep learning on images and videos [45]. While images can be flattened to a one-dimensional vector to process it with simple convolution filters, layers or even with recurrent and simple linear models, video processing and learning arrives with a bigger challenge. Fortunately kernels for convolution are able to have n-dimensional shapes though getting expensive in computation and memory consumption. The same issue occurs naturally with video transformers [46]. These follow analogously the already presented architectures in text and audio classification as they are transfering the researched benefits from transformer models to the vision domain. Nevertheless they scale quadratically with respect to the input length. Thus further optimizations on the input which will be presented in the following, are helpful and even more necessary.

Slicing offers different options to reduce the input size from the video with a variety of selection methods [47].

- **Time Exclusive Slicing:** As the most intuitiv approach, time exclusive slicing picks one or more single frames out of the given video, resulting into a more compact version like a flip book. This selection can be done randomly or with a fixed selection of the frame indices [48].
- **Spatial Exclusive Slicing:** This method divides each frame into a grid of spatial patches. Then it chooses one of the grids (mostly the center) and picks the same patch over the full time dimension.
- **Cubicles:** Cubicles or tubelets [49] are extracted by choosing spatial patches in combination with its temporal neighbours. This is similar to spatial exclusive slicing but with more variety in the choosen grid positions.
- **Shards:** This method [50] is a combination of spatial exclusive slicing, cubicles and time exlusive slicing, where from random or fixed indexed frames the spatial patches are choosen.

Overall they reduce the input size of the input video to a fraction of the original size on the basis of the assumption that there are unnecessary parts in each frame like the corners and that videos with more frames per second are icluding more redundancy respectively to very small changes and movements between each time step. Furthermore the basic spatial exclusive slicing is not well suited due to the lack of removing the

described redundancy. In general spatial patching can be challenging, if the person in the video is moving or the main actors are in different position in each data sample.

Further methods for input optimizations regarding to the idea of tracking movements are researched like utilizing temporal difference images [47]. It follows the assumption that deep learning architectures are learning the features only from the moving part of the face and that non moving parts are unnecessary. Intuitive according to Ekman and the presented action units from Chapter 2.1.4 this hypothesis has a good foundation. With time exclusive slicing they picked frames both randomly and with a fixed distance between the time steps and subtraced the frames from each other to create a temporal difference image for each video. Each of them shows the moving action units which were present in the data sample before. Thus it shows the action units which are normally learned by the first layers from an architecture.

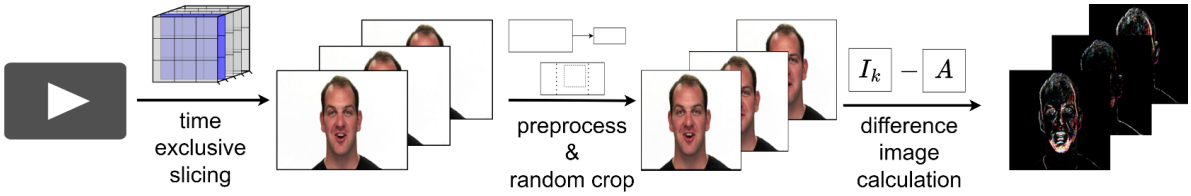


Figure 2.12.: An overwiev of the preprocessing pipeline [47].

This optimization method is shown by Figure 2.12 where the resulting temporal image shows only the moving parts from the video. Thus it may save some computation time but mostly it is able to save memory cost due to the reduction from 30-120 frames per input to only one temporal frame.

A similar but yet different approach is called optical flow [51], where for each video the moving parts and how they change is calculated and represented as multiple feature vectors. Here a video is represented mathematically by $I(x, y, t)$ where t denotes the time dimension as the single frames. The coordinates x and y belong to points defined in advance at which the movement is then tracked. These can be for instance the corners of the mouth, parts of the eyebrows or the eyelids. More points can be used to get a more precise resolution but also a larger input sequence. An appropiate amount would be between 20 and 30 to capture all 26 basic action units and some more details. The resulting optical flow while tracking is denoted by

$$\frac{\mathrm{d}t}{\mathrm{d}x} = I_x u + I_y v + I_t = 0 \tag{9}$$

where (u, v) representing the direction where the pixel at (x, y) is moving to. The resulting flow matrices can be used for a convolution deep learning architecture to determine the corresponding emotion. All in all there are these approaches to reduce the memory and computation time problem. If these aspects can be neglected a simple 3D convolution network will cover the time dimension and can be adapted to the structure of a video [52].

Unsupervised Classification Approaches

The challenge for unsupervised classification is the feature extraction from the video which needs some kind of tracking like optical flow. Thus it has to determine which action unit is activated leading to a combination as presented in Chapter 2.1.4. The already shown preprocessing possibilities from supervised approaches are visualizing a first look about possible feature extraction, which can also be helpful for unlabeled datasets.

Appearance-based Approaches are following the idea that different facial regions are defined by unique colours and textures. In general it is approached by mapping a coordinate grid absolute to a frontal picture of a head [53] or relative to fixed marks [54]. Each face region is divided into non-overlapping patches and blocks. Each block is represented as a single feature vector and contains one facial landmark which can be afterwards clustered to an action unit. Thus it observes the moving parts by colour, texture and the position in the frontal face it is sensitive to changes of light and even smaller changes of the head position. Therefore it is a suitable method for testing and methodological improvements but is not quite suitable for real life application due to these sensitivities.

Geometry-based Approaches are not well researched due to extracting and obtaining facial point locations [55]. However somewhat similar to the appearance-based approaches the idea for geometric feature extraction also looks for changes in the static face but more for deplacement of the same point location due to muscular changes. In combination with the approach from before it can result in improved performance independent from lighting and outer conditions.

Motion-based Approaches are based on dense motion rather than the discrete motion from facial landmark points as presented in geometry-based approaches. Image substraction, as one subset, which was already shown during input minimation for supervised approaches can also be used together with a combination of basic clustering algorithms like k-means [56]. Also the presented optical flow method is capeable to be used for unsupervised learning adapting it to quadtree decomposition for matching the important moving parts.

In general these feature extraction approaches uses k-means with an alignment to time series. Often used is dynamic time warping, which is a distancy metric and useful for time series with different lengths. Thus it has a drawback due to not always satisfy the triangle inequality which is normally an important property for these metrics. Further optimizations lead to a method called dynamic time alignnment kernels [57] defined by

$$\tau = \max_{Q} \sum_{c=1}^{l} \frac{1}{n_x + n_y} (q_{1c} - q_{1c-1} + q_{2c} - q_{2c-1}) \kappa_{q_{1c} q_{2c}} \tag{10}$$

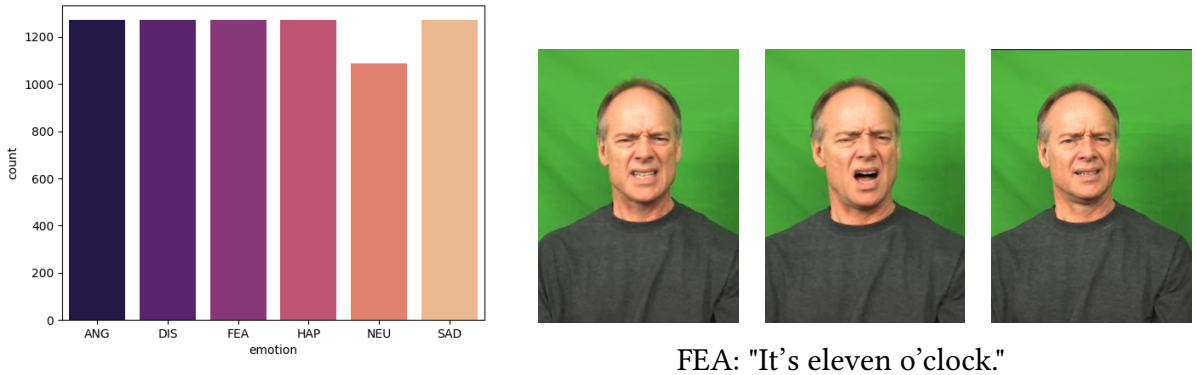
where $\kappa_{q_{1c} q_{2c}}$ represents the kernel similarity between single frames from two video inputs. A possible kernel could be $\kappa_{q_{1c} q_{2c}}(x_i, y_i) = \exp\left(\frac{1}{2\sigma^2} \parallel x_i - y_i \parallel_2^2\right)$ with σ^2 as the average distance to the 10% nearest neighbours [58]. Clustering the features is not limited to k-means but is the most common approach. Thus the research in time series clustering has developed over the last years bringing different metrics and algorithms to the front and can be adapted with little to no effort [59].

2.3. Datasets

For the multimodal analysis and the supervised approaches in learning a dataset consisting of at minimum video with audio is needed. From this state the text transcription can be retrieved during preprocessing. During this thesis, two different datasets are used which fulfill the requirements. The following datasets are based on the Ekman emotion model with some changes.

CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset) [8] consists of 7.442 clips from 91 different actors with their facial and vocal expression. It covers every of Ekmans emotions: *Anger (ANG), Disgust (DIS), Fear (FEA), Happy (HAP), Neutral (NEU), Sad (SAD)* with the exception of *surprise (SUP)* , because it can relate to any other emotion with a rapid onset. Additionaly there are a few annotation to the intensity of the sentiment, but it is only a small percentage regarding the amount of data. Therefore it wont't be used for further evaluation in this thesis.

They recorded twelve different sentence for each type of emotion like "Don't forget a jacket." or "I wonder what this is about." which are semantically neutral connotated. These snippets are between 50 to 80 frames long while it is recorded in 30 frames per second.

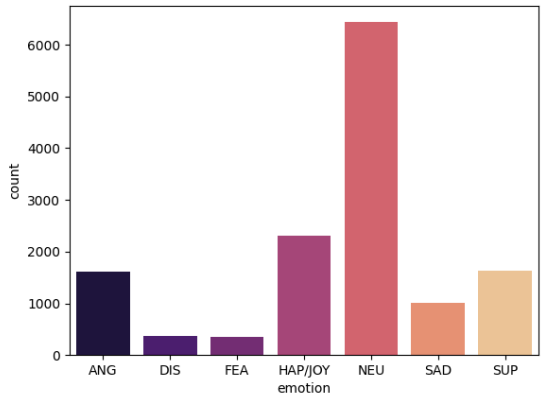


FEA: "It's eleven o'clock."

Figure 2.13.: Distribution and example of the emotion presented by the CREMA dataset

As shown in Figure 2.13 the distribution of labels is uniformly distributed with the exception of neutral. This won't leed to any problem, because the actual sentiments shown in the visuell data are very strongly acted so there are clear features available.

MELD (Multimodal EmotionLines Dataset) [60] is an annotated collection of short sequences from the series "Friends". In difference to the CREMA dataset or most other multimodal datasets for sentiment analysis like IEMOCAP [61] the given data has many multi-party conversation where there are more then one participant active and present in the most parts. This leads to a more complex and challenging feature extraction but approximates more closely the reality.





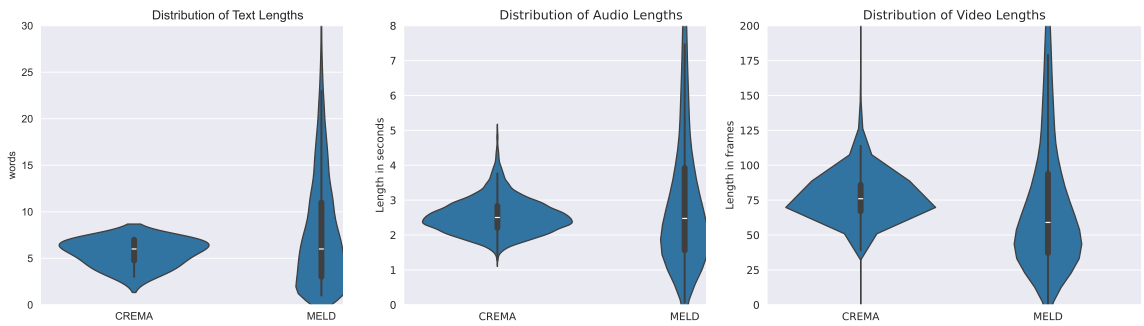
JOY: "You are a genius." (Chandler)

Figure 2.14.: Distribution and example of the emotion presented by the MELD dataset

Figure 2.14 shows the uneven distribution of the different sentiments, which can lead to an imbalanced training. Nevertheless it is a quite large dataset with 13.708 annotations which includes in addition to the emotion, the character as the speaking person which can be used to analyze the different features depending on gender or age.

Data Statistics and Analysis

Looking at the data distribution presented by the violin plots in Figure 2.15 a clearly pattern can be observed while taking a focus on the length distribution of text, audio and video. Wider areas along the x-axis are generally showing a higher occurence of the length. Further the range is shown by the y-axis. The CREMA-D dataset includes many shorter sequences with little to no scattering leading to an average of about 2.5 seconds of content. In contrast includes the MELD dataset a variety of different length upto more than five seconds material thus showing a mostly similar average of time. In conclusion CREMA propagates the lack of variety in sentences as shown in (a) through the video (c) and audio (b) content as expected due to their limitation to only twelve different sentences. With the more real-world applicability MELD features a high variety which might lead to the need of more data preprocessing like padding and a larger model architecture depending on the input size.



(a) Text (b) Audio (c) Video

Figure 2.15.: Distribution of the different modal lengths for each dataset.

Challenges

As already described before the datasets are bringing different challenges and requirements to the model architectures. There is also one obstacle for the evaulation later which comes from the CREMA-D dataset. It consists of the twelve sentence which are actred in each emotion. Therefore it is not possible to predict any relieable label for the emotion detection in text. A general challenge for the MELD data is the imbalance between each label. As the neutral emotion is by far the most common annoation the models will mostly predict only this category but will however achieve a high accuracy. Furthermore due to the imbalance in the actual data sizes and as before explained with Figure 2.15 the MELD dataset covers a high range of different input lengths for each modal. Therefore some adaption to the maximum input length for both datasets is needed or some kind of input compression so that in the end all data entries have the same length for training.

Chapter 3.

Related Work

Emotion detection is a research field which has developed much over the time. Also the comparison of multimodal emotion detection approaches and their results crystallized in the last few years. In the following chapter multiple exploration are presented with their idea and the result. Also a comparison of the different approaches will take place mostly while presenting the ideas shown in the works.

3.1. Audio and Video Based

Temporal Multimodal Fusion for Video Emotion Classification in the Wild by Vielzeuf et al. [62] builds a framework on audio and visual features where it tries different combination and different scoring of the features. They use convolution models as a deep learning architecture and the OpenSmile toolkit as an audio feature extractor to retrieve supervised the six emotions from Ekman and in addition the neutral state. The visual information is extracted frame by frame and as a consecutive video whithin two different models. For training they used the Acted Facial Emotion in the Wild dataset [63] which consists of different movie scenes and is comparable with the presented MELD dataset. In total they have an accuracy of 58.8% with the combination by a weighted mean but with no information about how they set the weights.

Multimodal mixed Emotion Detection [64] presents a similar method and implementation to the work presented before. Thus they create features from hand gestures and body movement aswell which are then forwarded into a support vector machine. To extract information about the visual and audio data they used premade frameworks like OpenEar similar to OpenSmile and FaceAPI which allows better tracking for faces and head rotations. Additionaly they used an infrared sensor from Microsoft for the gestures and moving parts of the body. Furthermore they classified the six Ekman emotions like the paper before but without any neutral emotion. As results they presented an accuracy of 77.5% while using data from the hand gestures, 65.2% for the body modality and 94.3% for using information from the head region. In total by combining all features they got an accuracy of 96.6% which is quite promising. Thus they are using the IEMOCAP database which is more comparable with the CREMAD database and brings more acted emotions into training than a more natural dataset would do which was used in the work before. Also they did not include any feature combination with weights or other methods. The high difference in the final accuracy shows the difficulty with natural moving parts or moving camera perspectives like shown in movie scenes.

3.2. Audio, Video and Text Based

Multi-level Feature Optimization and Multimodal Contextual Fusion for Sentiment Analysis and Emotion Classification [65] is using text, audio and video as input, extracting the features with evolutionary computing based models like GA or GGA [66]. They propose a multi layer feature extraction which propagates the raw input through different layers. The first subsets of information and patterns are extracted by unimodel GAs and GGAs, and then forwared into an recurrent LSTM network to retrieve more contextual input. This procedure will be repeated multiple times and between each step a fully concatenation of all features is applied and a split from unimodal to a trimodal architecture. For training and testing they used the IEMOCAP database as a static dataset and the CMU-MOSI data. This set consists of opinion videos from YouTube splitted up into multiple emotionally affected segments making it compareable with the MELD or the WILD database. As shown in Table 3.1 they got quite reliable results showing that a combination of the multimodal features can improve the accuracy. In single feature usage they got a score of 69.49% for video, 77.28% in text and 68.28% while using the IEMOCAP dataset and similar results for the CMU-MOSI model.

Modality	Accuracy CMU-MOSI	Accuracy IEMOCAP
T+A	77.06%	78.85%
T+V	76.65%	79.12%
A+V	70.00%	65.44
T+A+V	80.38%	79.52%

Table 3.1.: Comparison of results. A = Audio, V = Video, T = Text.

To combine the features they used the contextual fusion where the model trains the optimal weights for the composition. Furthermore it shows that there will not be much improvement between text and audio in turn of the same features combined with video features. Thus it presents a relieable architecture for both different types of datasets.

Multimodal Sentiment Analysis using Hierarchical Fusion with Context Modeling [67] follows a slightly similar approach like the work before. Instead of GGAs for feature extraction they used basic convolution networks trained with the IEMOCAP and CMU-MOSI databases. Thus for audio OpenSmile was used as a pretrained processor. They present a mostly identical feature combination strategie with a fully connected layer and gated recurrent units are used for contextual learning and to combine the features while learning the optimal weights. Overall they achieved a little less accuracy than the feature extraction with generative models but for text, video and audio in combination they got a score of 80.0%. Audio and video achieved 58.8%. Bimodal combinations with text got an accuracy of round 79% leading as well as the insight that there is not much room for improvement in comparison to the trimodal result. For the single model evaluation, text performed the best with an accuracy of 76.5%. The other two models bring about 50% to the evaluation which is quite low in comparison to the already presented methods.

Seq2Seq2Sentiment: Multimodal Sequence to Sequence Models for Sentiment Analysis researched by Pham et al. [68] presents an architecture based on a special class from recurrent networks. The basic idea is to first train a sequence-to-sequence model for two modalities and afterwards a combination of the already trained model and the third modal to the final result. Therefore they use the CMU-MOSI dataset and for feature extraction they build on different pre-trained encoder. For text input a GloVE [69] which represents its own idea of encoding besides skip-gram or bag-of-word models thus they used an already pre-trained one. COVAREP [70] was used to extract multiple important features and feature representation from the audio signals and for videos an motion-based representation extraction method is proposed. Altogether they did not measure the accuracy for single modal training due to the structure of the architecture. Thus they got an accuracy of 75% for all three modalities in combination. Interesting to note are the accuracys for the bimodal detections. Text in combination with video got 78%. The other two modal permutations lay at 44% for text with audio and 55% for audio and video. They discussed these results with the reasoning that the model architecture did poor training on video and audio features and recommend other encoding mechanisms and a bit different model architecture basing on sequence-to-sequence for future work.

Tensor Fusion Network for Multimodal Sentiment Analysis [71] presents the method of tensor fusion for the necessary feature combination in context of the multimodility. It describes an untrainable algorithm which attemps to combine multiple tensors by a reduction to a single tensor. This results in an interplay between a recurrent and LSTM based model for each modality and the proposed tensor fusion layer. Only for text input they showed an own method to extract the necessary features also by combining LSTM cells and linear layers. Like the other works before they set their foundation for audio extraction on pretrained models like COVAREP but they combined multiple processors for different features as peak slope parameters or other high level audio information. For video they also refer to FACET as the facial expression analysis framework able to extract up to 20 action units. On to the tensor fusion layer some fully connected deep layers will learn from the combination tensor to predict the emotion. Overall they reached with their unimodal network an accuracy of 77.1%. Also here the baselines representing every single modal are bringing a high accuracy for text processing by 74.8% and lower precision for visual and acustic features with round about 66%. It outperforms the basic convolution network proposed by the first work and reaches good performance in relation to all other presented works thus being a more rare approach while using an untrainable feature combination.

A Social Emotion Classification Approach using Multi-Model Fusion [72] implements a hybrid form of convolution and recurrent networks to build a context based extraction method for all three modals. For video they combine LSTM and 3D convolution to a proper time understanding model and a simpler CNN-RNN architecture and a bag-of-words for text classification. Again for the feature extraction from the audio signals they proposed OpenSmile as a relieable framework. For the subsequent feature combination they introduce multi-kernel learning [73] which retrieves the result from each kernel gernerated by each modal within the CNN architecture. For comparable results they used the IEMOCAP database aswell as the MOUD dataset. It is like CMU-MOSI a collection of YouTube videos and snippets and therefore a more real-world illustration but it has over 300000 data snippets which makes it in comparison to all presented datasets really large. Therefore they got pretty relieable results

and a trimodal accuracy of 95.88% with the MOUD dataset and 78.75% for IEMOCAP showing that a bigger dataset can lead to more exact results.

All in all the current works in the field of multimodal emotion detection are not showing much differences to each other. As shown in the following Table 3.2 the main dissimilarities are lying in the basic training architecture to train the feature combination. Thus there are mostly some kind of reccurent architectures deposited to keep track and of context bringing an important focus. Furthermore most of the current works are bringing some indistinguishable accuracys but showing that a bimodal combination of text and audio brings a nearly good result in comparison to the unimodal architectures. Also the usage of pretrained processing layers and feature extraction like FACET, GloVE and OpenSmile is highlighting which servers a huge reduction in computation and necessary preprocessing time. It should be noted that there is no research about the multimodal emotion detection featuring the leading transformer architectures which could be found.

Author	Year	Dataset	Architecture	Fusion Method	Accuracy Bimodal	Accuracy Trimodal
Vielzeuf et al. [62]	2017	WILD	CNN	Weighted Mean	58.8% (A + V)	-
Majumder et al. [64]	2018	IEMOCAP	SVM	Linear Layer	96.6% (A + V)	-
Huddar et al. [65]	2020	IEMOCAP, CMU-MOSI	GGA, GA, Recurrent LSTM	Contextual Fusion	77.06% (T + A) 76.65% (T + V) 70.00% (A + V)	80.83%
Majumder et al. [67]	2018	IEMOCAP, CMU-MOSI	CNN	Recurrent Networks	77.30% (T + A) 77.80% (T + V) 58.80% (A + V)	80.00%
Pham et al. [68]	2018	CMU-MOSI	Seq2Seq	Hierarchical Fusion	44.00% (T + A) 78.00% (T + V) 55.00% (A + V)	75.00%
Zadeh et al. [71]	2017	CMU-MOSI	LSTM	Tensor Fusion	-	77.10%
Xu et al. [72]	2020	IEMOCAP, MOUD	CNN-RNN	Multi-kernel learning	85.36% (T + A) 92.37% (T + V) 91.14% (A + V)	95.5%

Table 3.2.: Overwiev of the compared works showing the bimodal and trimodal results. A = Audio, V = Video, T = Text, - = not provided / not evaluated.

Chapter 4.

Design and Implementation

Given the six emotion from Ekman explained in Chapter 2.1.1 and a neutral label, the following chaper focuses on the implementation of the different supervised model architectures for each modal. This will be done for the presented datasets MELD and CREMA-D which are both challenging different aspects of emotion detection. Furthermore the important preprocessing and feature extraction for each model are described. At least the bimodal and unimodal combination are implemented showing different types of feature combination or voting and scoring methods. Figure 4.1 shows an overview about the following implementatio goal. In the following each component of this architecture will be described and discussed.

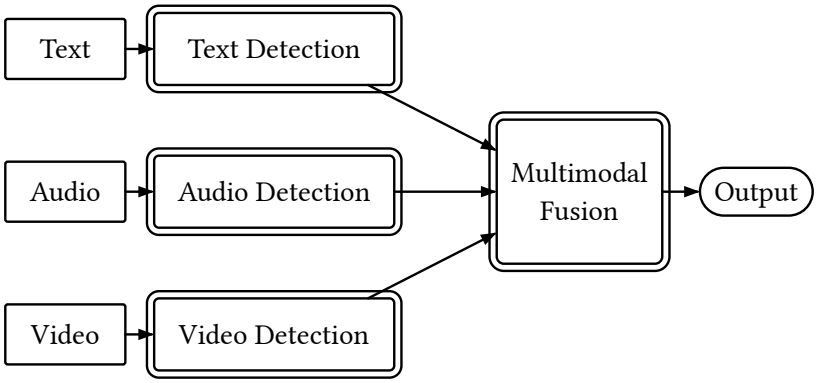


Figure 4.1.: Pipeline for multimodal emotion detection.

4.1. Unimodal Text Classification

As described in Chapter 2.2.1 transformer models are quite suitable for text classifcation. However they need some sort of word embedding for preprocessing. Fortunately already existing Large Language Models are capeable to process text as normal conversation input for giving a mostly correct answer to the prompted sentences. Also the most common models are also pretrained on very large datasets for example paper from scientific websites or collections of sources and articles like Wikipedia. Thus needing some adapations and further finetuning to the used datasets in this work, the following three state of the art pretrained networks are used for measurement and classifcation.

Llama3 [74] developed by Meta presents a generative deep-learning modal based transfomers. It consists of over 128 thousand known vocabularys from open sources in the internet.

With more than four billion parameters and its well preprocessed dataset they show relieable results in text conversations. As a natural language understanding model similar to ChatGPT or other various models. Llama provides contextual understanding in a chat like conversation with the user. In contrast to the most big architectures, Meta decided to publish their model open source with all the pretrained weights and different interfaces to run the product locally.

Multi-EmoBERT [75] was originally applied to detect fake news with the support of emotion classifcation. Thus it is more an architecture than a pretrained model, bringing the possibility to train and finetune the architecture with own text data. As already briefly explained in the background of text preprocessing BERT is a bidirectional encoder with an underlying transformer developed by Google in 2018. The multiemotion aspect gives multiple labels and their probabilities for a given input leading into a higher variety and more options to choose the correct label.

T5 Model [76] is a text based transformer model which settles on text-to-text transfer learning, following the idea from Llama 3 and equivalent also trained on open source crawled data from the internet. The advantage to the already presented architectures and implementations is an already existing open-source finetuning on sentiment and emotion detection reducing the effort for necessary preprocessing.

Architecture Overview

As said an emotion classifcation with text will only be done with already pretrained models thereore they will be considered as a black-box but with some options of finetuning.

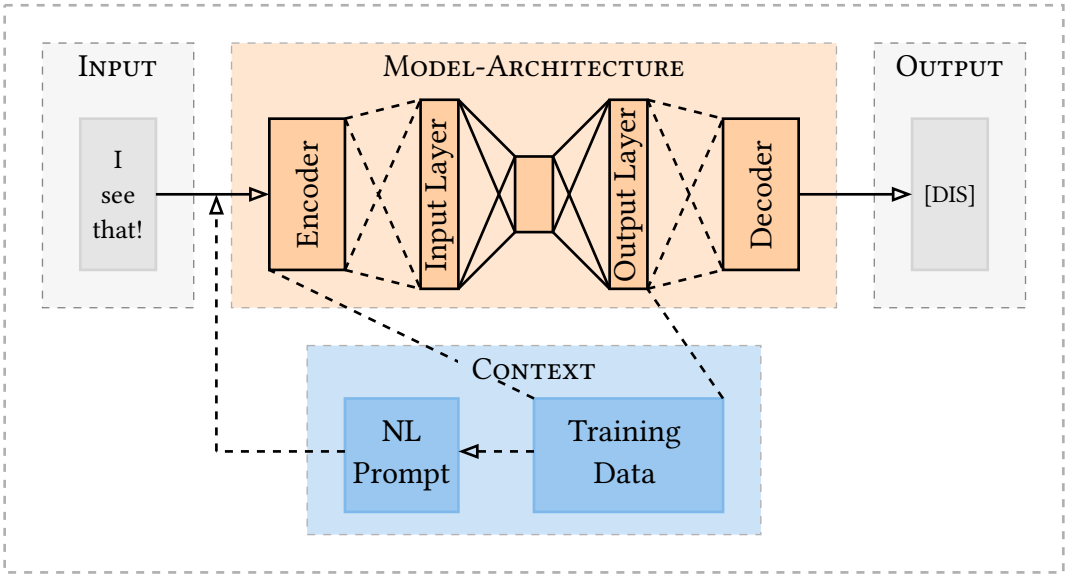


Figure 4.2.: Structural procedure for emotion classification and possible influences on pretrained text-to-text models.

Figure 4.2 shows the sequence of the processing which can be adapted to the three shown transformer models. In general such models are needing an encoder with a tokenizer and decoder to understand the given natural language input further to output human understandable labels for the corresponding emotion. In between, their model architecture is given mostly by

their adaption of a transfomer and is already well trained which already results in a set of predefined weights. For finetuning these models are offering in general two different methods, prompt engineering and post-training tasks, which allows adding context to the architectures or weights. Some text-to-text models are trying to simulate human-like chat conversations with contextual memory over a talk. Foremost it gives the opportunity to define tasks which will be fullfiled by the artifical opposite. Also it introduces a scope to feed in the available training data to the model by embedding it into the chat prompts and define the task to learn temporarily from it. This introduces the problem that the given knowledge is just temporarily available in the scope of the current chat window and afterwards permanently deletet and must be repeated every time. Therefore the second method for finetuning shows a procedure to train on top of the current pretrained weights and the encode layer with a model specific interface. It offers a more relieable context injection in comparison to the prompt engineering method by giving more transparency about the learned content.

4.1.1. Preprocessing and Prompt Engineering for LLM

Model	Response	Type of Response
Llama3	I'm glad I could share the excitement with you! It's always thrilling to spot a clever joke or play on words. If you're ready for more, I've got a few more where that came from!	Natural Language Conversation
Multi-EmoBERT	anger: 0.0098 disgust: 0.0050 fear: 0.0113 joy: 0.3621 sadness: 0.0121 suprise: 0.0564 anticipation 0.5360 optimism 0.4288 pessimism 0.0121 trust 0.0330	Probabilities for every emotion
T5 Model	joy	emotion label

Table 4.1.: Answers to the prompt "I see that!" from the three text-to-text models without finetuning and any context.

Fortunately all of the three given models are implementing an own language understanding system. Therefore it is not necessary to postpone and train an own word embedding network. Mostly they rely on some variants from bag-of-words to encode the vocabulary tokens. In all three cases the models consist of an encoder and a decoder layer for the text-to-text understanding and their specific transformer based architectures to give them their expected user-

oriented functionalities. Thus just input a sentence to retrieve the emotion label won't work well in the most cases due to the attemp to imitate real-world conversations. Looking at the results for the input sentence "I see that!" the models are showing the following responses in Table 4.1.

Llama 3

Llamas response shows the conversation like answer leading to the need of some finetuning. Fortunatley the model has contextual memory which can be used to predefine a task with natural language for the architecture. In this case the chore with an example for better understanding is given and repeated after every prompt to refresh the contextual memory. Further as shown in Figure 4.2 and explained before the train data can be embedded in the progression of the conversation to give the model a higher overview over the given data which can lead to a higher accuracy.

A possible prompt which is used later for the evalation is:

> "Giving the following seven emotion with their abbrevation: Anger (ANG), Disgust (DIS), Fear (FEA), Surprise (SUP), Sadness (SAD), Happiness (HAP), Neutral (NEU). I'll prompt you a sentence and you will answer with the corresponding emotion abbrevation. For example I'll ask: "My day is awesome" and you're answer would be "[HAP]". Don't forget the brackets. Sentences, or questions like "okay" or "lets go" are also sentences where you have to guess the emotion. Alright?"

With this task the model outputs a more sensible answer in the right format for the same example sentence. This prompt can be changed slightly to feed in the contextual information about the training data without much effort.

Pseudocode

Multi-EmoBert

However as mentioned it is a bit more complicated with the Multi-EmoBert due to the lack of any context based chat functionality thus they already give the emotion label as a response. Looking back at Figure 4.2 and the notes before there are possibilitys to train the architectures in particular the encoder and the weights on top of their pretrained values with an own set of data. This opens up the opportunity to finetune the models and enhance the results. Simply the data from the training dataset has to be processed to the specific format which is already offered by an interface from the source code. Further an application of a threshold to validate the correct emotion label is necessary as it outputs all probabilities for every sentiment showing another variable factor as shown in Listing 4.1 which has impact on the accuracy. A lower value might increase the chance to include the correct emotion in the set of all sentiments which are guessed with a higher possibility. Thus the amount of false positive emotion must be considered as well. With a higher threshold the possibilities to guess the correct label

might be lower but also the magnitude for all misclassified labels is lower. If there aren't any classifcation which reached the minimum possibility, the sentence is tagged as neutral. All sentiments given by the model which are not part of Ekmans six basic emotions or neutral are ignored because they are based on the original training data and have much lower possibilities after retraining the model with an own dataset excluding these additional sentiments.

```
def choose_emotion(P, th):
    L = [ ] # list for output

    # loop through every possibility
    for e in P:
          if P[e] >= th: # if matches the threshold
               L.append(e)

    # if none of the emotion reaches the threshold
    if  L.is_empty():
          return ["neutral"]
    return l
```

Listing 4.1.: Simple algorithm for choosing all emotions from the possibility-space P with an threshold th.

Pseudocode

T5

The T5 accepts both methods for finetuning thus prompt engineering does not seems to be mandatory due to the already labeled output. Thus the combination might show interesting results in comparison to the other two architectures with only one of the two finetuning methods. In comparison to the Multi-EmoBERT model it will not offer a direct interface for dataset conversion and an active method for training but it has an low-level connection which allows training like a normal model but with respect to the pretrained weights. In addition the documentation gives some functions to enable regular training. As a PyTorch submodule it includes a loss function based on the cross entropy loss function to compare the resulting token values. Also the encoder and tokenizer part is able to recognize new tokens and adapt itself without any further external influences or preprocessing.

Pseudocode

4.2. Unimodal Audio Classification

In contrast to text classification there aren't many pretained and large networks available. Thus many different architecture and model types exists as described and shown in Chapter 2.2.2. Fortunatly multiple feature extraction methods and tools are useable bringing a large range of different information on the basis of fourier transformation. Furthermore an own encoder and decoder mechanisms must be implemented for audio understanding and reliable labeling.

Preprocessing and Feature Extraction

OpenSmile extracts a variety of valuable information from an audio segment giving a good overview about necessary components which are useful for the corresponding set of emotions. With the knowledge from Table 2.1 a small cutout from all extracted features is presented in the following. This excerpt shows the significant types of information which are mostly supported by similar features.

Loudness describes the volume of an audio and is able to describe firmly two different sets of emotion which are more calmer sentiments like sadness or neutral and more powerful ones like anger or happiness. In general louder voices leading to more agressive emotion while calmer audios are representing harmonious or more melancholic sentiments of a person.

Alpha Ratio shows the relation between frequency above and below 1000Hz. In support of loudness, this ratio analyizes vocal volumes and not only the volume of the audio sample itself. Also some information about timbre from the data can be obtained.

Hammarberg Index [77] can be briefly described as the intensity difference between the maximum intensity of the signal in a lower frequency set and the maximum intensity in higher frequency thus giving an energy distribution measurement for the given audio. Similar to loudness the energy index normally can be mapped similarly to the two sets of emotion.

Slope measures how fast a given audio signal tries to reach high frequencys leading to different possible assumptions about pitch changes and pitch ranges which are important for several types of emotion like fear or disgust, as shown earlier in Table 2.1.

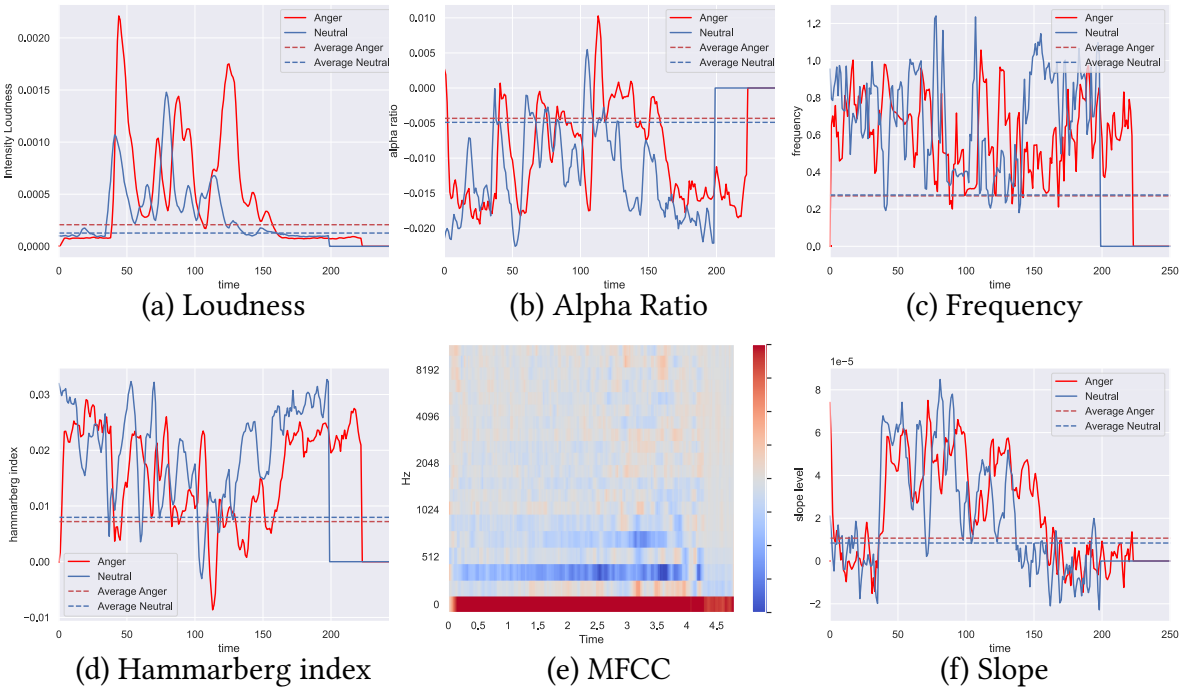
MFCC (Mel Frequency Cepstral Coefficients) are a compact representation of frequencys where different pitches are visible. They are often used for speech synthesis and other detection methods due to their high information value. In general features like the speech rate which is important for emotion as for example fear or sadness can be obtained.

Jitter is a loss measurement while working with digital audio signals and shows if there is noise in the data while comparing frequencys and oscillations. However it is in this case not necessary for emotion detection thus it is generally helpful for noise detection in this signal processing.

Frequencys are a low-level measurement for the feature extraction. While MFCC are giving a compact overwiev, these extracted frequencys showing raw information about the audio.

Thus as already described earlier they are important for all kinds of audio classifcation and their information retrieval.

More features are also present resulting in a matrix with 20 different types of information which can be processed by different network architectures. Further these layers are including redundancy in different formats and representation for better learning and feature extraction within the models. A comparison of the subset of traits from the audio shows the changes in their data. Figure 4.3 shows the presented excerpt from the features which are extracted by OpenSmile.



(a) Loudness (b) Alpha Ratio (c) Frequency

(d) Hammarberg index (e) MFCC (f) Slope

Figure 4.3.: Cutout from the different OpenSmile features in comparison between all anger and neutral audio samples from CREMA-D.

Each plot shows the changes of the metrics as a time series for the more powerful emotion anger and the neutral baseline. To retrieve this overwiev all audio data for each of the two emotions from the CREMA-D dataset are added together and the average was drawn to retrieve an average signal and to compute these features with OpenSmile. Visible in the loudness-plot the red lines show the visualization of anger. It is in general higher than the neutral plot which is connotated by the blue color. Furthermore the subfigure shows that fourious emotion are starting much louder in the main than the baseline data. This is underlined by looking at the average drawn out of the signals. Here also the main loudness from anger is much higher as from neutral emotion. As shown later it is also the highest difference between the averages leading to the assumption that loudness is a quite important factor and feature for the differentiation between these two emotions. Similarly a look on the alpha ration (b) shows a higher average from anger in comparison to neutral. As already explained this ratio shows also some information about the voice volume and the timbre which further supports the theory about the importance of loudness for this comparison. Interestingly the high amount of negative values of the alpha ratio are showing that there is more energy in the lower frequencys than in

the higher frequencys. In this case lower frequencys are the range between 50Hz and 1000Hz. To compare these values a male voice has generally a mean of 111.8Hz and 161.3Hz for females [78] showing the reason for this distribution and values in the ratio and is also supported by the distribution of frequencys (c). Even though it looks unequally distributed, the average for both anger and neutral is nearly the same. The very small difference might show a contrast between the sentiments like loudness or other high-level information due to its low-level representation where the most features are obtained from. More likely are small numerical issues for example instability in calculation of the averages or the difference can be canceled out with much more data. For this overwiev the CREMA-D dataset just offerend round about 1000 samples for neutral and slightly more than 1200 samples for anger out of the training set.

Normally expecting from the Hammarberg index anger must show more energy than the neutral emotion as already described in the alpha ratio. Thus visualized in (d) from the overwiev figure the average from neutral is a little bit but visibly higher than the calculated value from the other sentiment. This may result from different reasions related to the recorded voice quality and timbre. Sometimes more detailled voiced audios with less noise. Also a softer timbre might lead to these results. Finally the vocal pitch could have an influence and can result to these small yet existing and a bit unexpected difference.

Furthermore interesting is the MFCC plot (e) which shows the difference from anger and neutral emotion by subtracting each MFCC value stream from the other. Each colored cuboid represents the intensity of the mel coefficient measured in decibel. Transparent areas are showing that there is no difference between both values. Reddish parts describing that the mel coefficient for the anger emotion is higher than the coefficient for neutral. On the contrary bluish colored cuboids representing higher values for neutral voices. Looking back at the resulting plot in Figure 4.3 there is a high difference in the low frequency are in the range of 0Hz to 150Hz. As already described this is the interval where the average pitch for male voices lies showing the more active and relevant parts of the plot. This shows that anger voices are having a much higher mel coefficient than the neutral audios meaning that there is much more energy in the recording like loudness in opposite to the baseline. In the higher freguencys the mel coefficient in the neutral emotion is sometimes more present thus there is not many activity by the acting voice in the signal. The values might occur due to noise or compression artifacts in the recording data. While hearing into the audio samples a constant white noise can be observed which encourage this assumption. In a few times this sound could be the same in both audio categories leading to the colorless areas in the presented plot.

At last from the cutout the slope (f), shows similarly to the Hammarberg index and alpha ration a small but observeable difference between anger and neutral. These information about the pitch range and change shows that there are some importances within emotion classifcation and that there exists trainable features for the neural network architectures. Negative values are are usually an indication of pauses between words and sentences which can be important for measuring the pace of the speaking person or other information like spoken words per minutes.

Overall there are many more observeable features extracted by OpenSmile which are covering many aspects from audio signals. There are also differences visible between each emotion and the baseline neutral. However, some of these are quite small but in some ways short pat-

terns exist. Also as already shown there are some redundancies within the choice of extracted features which will not be a problem and might result in better training. All feature combination which are earlier shown in Table 2.1 are available and measured by OpenSmile and also shown in the given cutout from Figure 4.3. Loudness shows changes in the intensity, MFCC is important for the speech rate and articulation. With the alpha ratio the voice quality can be observed and slope shows features for the pitch average, range and with in changes. In every plot and feature there are many zero values visible. Generally they exists at the end of the signal series due to padding with zeros and are the reason for the sudden constant progression of the time series. A neural network wouldn't have any problems with these padded values and normally only tries to learn from the non-zero features.

Model Architectures

As already presented there exists a lot of different architectures which are able to learn and train from audio data. Most commonly they have some kind of recurrent and bidirectional neurons or in general some context understanding which is important for time series training.

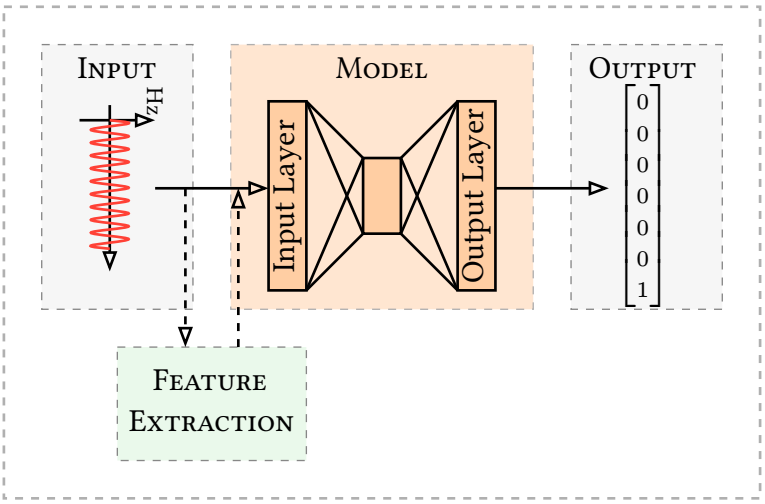


Figure 4.4.: Structural procedure for emotion classification in audio with a possible pre-trained feature extraction.

Similar to Figure 4.2 which shows an abstract visualization of the text to emotion processing pipeline, Figure 4.4 presents a small overwiev about the necessary components for the detection in audio signals. Starting with the input it is possible to train from it without any external feature extraction. Thus only the audio transformer which was presented earlier is currently capable enough for relieable results. In general the convolution based architecture mostly needs a predefined feature extraction for better and faster learning results. These two models are the main models implementet in this thesis for emotion understanding in audio. Each architecture will output the calculated logits which are representing the possibilities for each label to be correct. These are afterwards used to map them on to one-hot vectors showing the corresponding emotion as positive (1) and negative (0). This might also allow multi-label clasifcation but is not introduced further in this thesis due to the limitation of the scope. For the optimization task of the models the cross-entropy loss is used for each to calculate the error between the resulting and softmaxed logits with the given one-hot vector target. Table 4.2 shows

as an important overwiev the amount of trainable parameters for each model. Generally the three presented text models are implementing round about 4 billion parameters or more each, however they are already mostly pretrained. Both audio architectures are well comparable and within a reasonable range to these models allowing less noise due to higher model differences.

Architecture	Trainable Parameters
Convolution	2.176.474.377
Audio Transformer	ToDo

Table 4.2.: Amount of trainable parameters for both implementations.

4.3. Unimodal Video Classification

Due to the feature extraction methods in audio emotion classifcation which output multiple matrices with their corresponding values the detection in videos won't differ much from each other. Thus as already discussed the problem with the additional time dimension within the videos is still existing and needs some preprocessing for improvements and optimizations to reduce the computation time while preventing any avoidable loss in training and evaluation.

Preprocessing Methods

As already discussed earlier there are several methods and ideas for preprocessing the video samples before feeding them into the networks. The simplest idea but most expensive in computation time would be to just use the full length with all available frames. The CREMA-D dataset would support this idea without much performance issues as its highest amount of frames in a video is 125 and an average of 75 frames but looking at the MELD dataset shows the significant difference with more than 200 single images in one video. A possible way which is also easy to implement would be to just extract important parts of the videos to represent all most significant features and movements of the speaking person. In general this is hard to track and measure without any further extraction or classifcation of the material. Thus an easier way is to pick every n-th frame out of the given videos instead of every frame. This will reduce the memory costs by the given picking constant and reasons by assume that a video with about 60 frames per second must include a bunch of redundant information which are skippable. Due to the different lengths of the videos, this idea will generate different amount of important cutouts. Therefore padding with zeros would be important and in the average case there will be as many frames as the longest video divided by the constant would have, leading especially within the MELD dataset to many zero values due to the inbalanced distribution of the lengths. Adapting to this problem a further development would consist of a fixed amount of picked frames with different distances and intervals between each other. In this thesis 30 cutouts from each video are choosen randomly which should be enough to represent the actual content and information flow without much loss. Also this constant represents nearly the average video length divided by 2.

Premade feature extraction methods like FACET which are directly looking for information about the presenting face and acting persons in these videos are suitable for scientifically recorded datasets like CREMA-D where only one persons sits directly in front of the camera. Due to the usage of the MELD dataset, which includes more real-world near data and where the video cutouts showing sometimes additional persons in a snippet or where the actor won't sit right in front of the camera. These kind of data collections are not suitable for FACET or other premade feature extraction methods regarding videos.

In total there won't be much preprocessing besides randomly picking 30 frames out of each video. As usual the data will be normalized which will loose information about the color and just keeps the grayscale information in the cutouts. This will also forces the model to learn specifically features about the moving parts in the faces. Thus in this case every color is given by an seperate channel for each RGB (red, green, blue) value. This should not make much difference but can give the model some colored reference points.

Another problem which occurs before the video can be forwarded into the model is the actual size of a single frame. Generally for example the vision transformer model is tested with small videos and images in size of 16x16 pixel [79] which won't represent the given data from both datasets. Within CREMA-D all data samples have the resolution of 480 pixel width and 360 pixel height. This results into over 100.000 values per frame however in comparison to the MELD dataset it is still not as problematic. These videos in the set are much larger due to their origin as samples from a television show made for bigger screens. Therefore they have a resolution of 1020 pixel in width and 720 pixel in height which is nearly sevenfold the amount of values per frame and video in comparison to CREMA-D. The only solution for this problem is to crop the videos to a workable size without any loss of information. CREMA-D offers an realtively easy approach thanks to the static recording of the speaking person which is generally in the center of the video with some surroundings from the background which can be cutted off. Thus as already shortly presented in Chapter 2.3 there are several persons within the MELD videos besides the speaking actor. Also as shown in Figure 4.5 the position of the observation target differs from data sample to data sample which is marked red for better visualization.



Figure 4.5.: Example cutout from the MELD dataset with marked active speaker.

There are two necessary steps available to extract the active person automatically as shown in Figure 4.6 and as an example visualized in Figure 4.7.

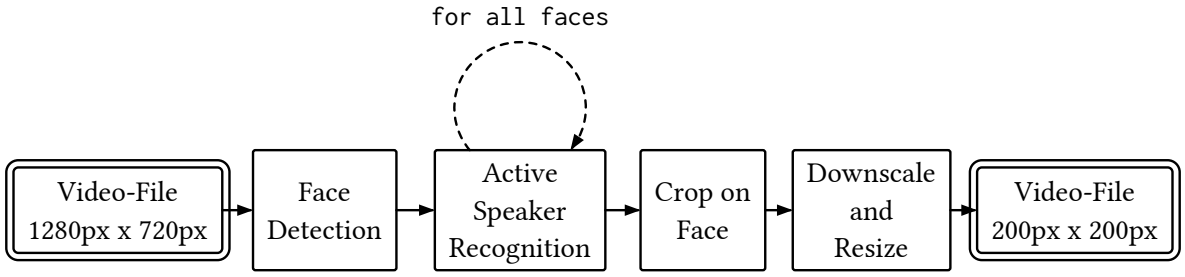


Figure 4.6.: Preprocessing pipeline for videos.

First all persons within the video have to get detected by an algorithm. Fortunatly computer vision librarys like OpenCV for python are able to recognize faces in images and therefore also in videos using pretrained machine learning algorithms. In the second step the active speaker must be found. There are also libraries which are able to analyze if the given audio matches to the lip movement from a person. Combining those methods all unnecessary parts like background actors can be cropped out of the videos which leaves only the important active part in a smaller video size.



(a) Face recognition. Red = Active Speaker, Blue = Background Character

(b) Cropped video after active speaker recognition

(c) Resized and downscaled video

Figure 4.7.: Example for the video cropping preproceesing pipeline.

At the end a futher downscaling is applied on both datasets regarding the relative face position using again the face recognition capability from the computer vision librarys. This leads to a video format of 200 pixels in width and height as shown in (c) from Figure 4.7 reducing the computation and memory cost by a lot. Thus there are sometimes some visible artifacts in the video which are resulting from camera shaking or unclear face positioning by the actors. Due to these issues the machine learning algorithms and librarys which are used are some times not as precise as the human understanding for face recognition. Also there might be some loss in video quality due to the downscaling which could lead to a depletion of subtle movements in the given face region. But with this preproceesing a model should be able to recognize patterns and similarities which was mostly impossible before.

Model Architectures

Similarly to the proposed audio architectures, the following structure for the classification in video won't differ that much from each other. Thus for better performance the preprocessing will be mandatory for the convolution based model and the video transformer architecture

while the OpenSmile extraction was optionally due to a possible understanding of the raw audio signal. The following overview in Figure 4.8 shows again as for text and audio, an abstract architecture about the necessary steps for audio classification.

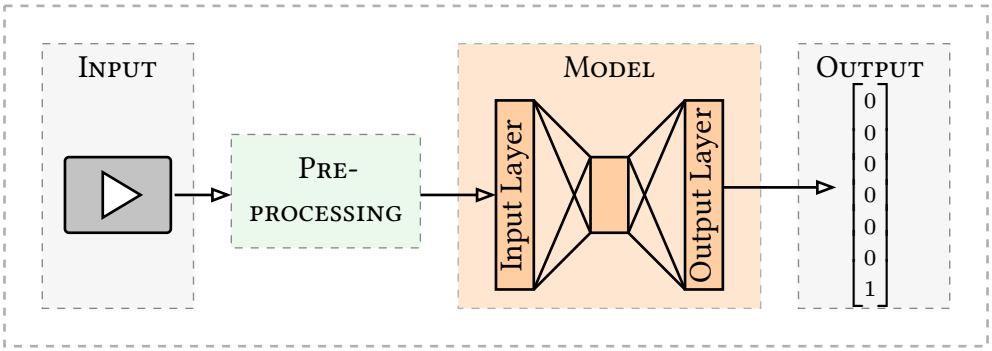


Figure 4.8.: Structural procedure for emotion classification in video.

As explained and proposed before the video will be preprocessed first where it is downscaled and cropped generally. Especially the data from the MELD set runs through the speaker recognition pipeline to filter out the most of the unnecessary parts from the samples. Also in both datasets the amount of frames is reduced to thirty with the assumption to reduce redundancy and to save computation cost. After the normalization step which is still part of the preproceesing the videos are feed into the model architectures. As the input has a size of $(\text{batch size}, 3, 30, 200, 200)$, describing the (batch_size, channels, frames, width, height), the convolution based approach uses 3D kernels to approach the trainings loop on the given batch regarding the additional time dimension. In contrast the audio convolution model only uses 2d convolution due to the dimensionality of the structure from these time series. Every block in the network consists of a layer norm, the actual weighted convolution layer and a max-pooling. After these blocks some linear dense connected layers are following for further downscaling of the remaining shape and the final mapping to the expected class shape which is in this case and also in the case of audio classificaiton (batch size, 7) representing each emotion as a one-hot encoding. This is also visible also in Figure 4.4 and Figure 4.8 while looking at the output block thus the model first outputs the logits as the likelihoods for each class. An argument of the maxima (argmax) is able to compute the corresponding label from the probabilities. Furthermore the transformer model is an adaption of a vision transformer which only accepts single images [80]. The implementation is currently open-source available[1] as an trainable pre-build architecture which only has to be integrated in the same training loop as for the convolution model.

Architecture	Trainable Parameters
Convolution	2.504.047.929
3D Vision Transformer	1.266.615.309

Table 4.3.: Amount of trainable parameters for both implementations.

[1] https://github.com/lucidrains/vit-pytorch

For comparison Table 4.3 shows the amount of trainable parameters for each architecture. Looking back at Table 4.2 the convolution model has a related size to the other audio based convolution model. Also in comparison to the pre-trained text models they are quite smaller but they should have enough parameters for a relieable result. The 3D vision transformer architecture features a lower amount of trainable points in the given model. Thus as proposed in the original paper, which introduced these kinds of architectures, they generally will perform similar or sometimes even better than traditional models with less parameters.

All models will be trained in twenty epochs with an 90/10 split for training and testing. This can give an overwiev about the efficiency of the architectures and also about the consistency and reliability of the results. Normally these models have to be trained over month for realistic and nearly perfect results but this will exceed the scope of this thesis work. Further the cross entropy loss is calculated for the gradient descent. Also the Adam optimizer is utilized for further optimizations in the calcaulation and uses a learning rate of $\mu = 0.00005$.

4.4. Multimodal Classification

Here the bimodal and trimodal emotion classifcation are unified as they won't differ that much in their structure and the methods for combining the outputs from every model. As the unimodal detection was straight forward a mapping from an input to the corresponding output some new ideas must be appended at the end of the given models. Some related works have already presented different feature combination methods like kernel fusion or other ideas. Thus in this case it is more difficult due to some limitations by the architectures which are presented in this thesis. It is too hard and nearly impossible to extract the kernel and logits from the pre-trained text classifcation models because they work like a blackbox where only the output which is always text can be considered. Also the both implemented transformer architectures for emotion detection in audio and video are not suitable for feature combination due to their lack of useable interfaces to extract feature information. These architectures will also only give their output logits. Only the two convolution models can give the required information for kernel learning or other feature combination. As a solution for these problems this thesis will only take a look on scoring methods as a simple form of mapping the three outputs to one final emotion. There are some different main methods which are suitable.

Scoring Methods

A scoring method is a way to weight multiple inputs by their importance for the final output. For example the first architecture is less crucial for the result as the second and third architecture. The task is to find a optimal function $f : \mathbb{V}^{n \times m} \rightarrow \mathbb{V}^n$ where n describes the amount of emotion which are able to get classified and m counts all possible inputs. In this case $n = 7$ for the six base sentiments together with neutral and further $m = 3$ for the three given modals.

Neural Networks which tries to learn its own weights are a good design for the simplest approach for scoring without further complexity. Here a single linear layer is used with the predictions of all training data and the actual output. After the learning phase this model should have an internal weight system for the multimodal input. As they try to solve an op-

timization problem by minimizing the loss they optimize the function f analogous to this. A drawback of this method shows the blackbox-like behavior of neural networks. After training and testing nobody is able to tell which input was important for the final result and which was not necessary or just a small influence. Also the supervised model architecture might be a drawback in comparison the the following unuspervised approaches.

Brute-Force is more elaborate method regarding time complexity. The idea is to permute all possible weights for each input. Looking back at $f : \mathbb{V}^{7\times 3} \rightarrow \mathbb{V}^7$ it can be simplefied and adapted to feature a weightened and parameterized function

$$f(a, b, c) : a\mathbb{V}_0^7 \oplus b\mathbb{V}_1^7 \oplus c\mathbb{V}_2^7 \tag{1}$$

where $a, b, c \in [0, 1]$ are denoting the different weights for each output and $\oplus$ the arithmetic for combining the vectors. The goal is to minimize the difference between the actual target $\hat{f}$ and the calculated output.

$$\min\left(\hat{f} - f(a, b, c)\right) \tag{2}$$

As already mentioned the brute force algorithm will permute all weights with a step size of 0.01. This results in a total of $101^3 = 1.030.301$ permutations. Reducing the step size will increase the amount of possible test cases cubical which underlines the high time complexity. Also it might not be able to find the most optimal solution globally but just locally in the permutation space. To find the global solution the function has to leave the discrete numeric space which is impossible for machines.

Major Voting is a strategy which only works with more than two different inputs. The main idea is quite simple: the most voted emotion must be the corresponding emotion. For example if the input is $(\text{sad}, \text{sad}, \text{neutral})$ the calculated output should be "sad" due to its major occurence in the space. This is also a more simple method and won't solve the optimization problem. Also there is no differentiation between the input models.

```python
def major_vote(L: list[int | float | str]) -> int | float | str:
    d = { } # dictionary for counting the elements
    for i, e in enumerate(L):
        if not e in d:
            d.update({e: 1})
        else:
            d[e] += 1
     # sort by value, highest value at pos 0
     return sorted(d, key= d.get, reverse=True)[0]
```

Listing 4.2.: Python implementation for the major vote algorithm.

This approach will only focus on the output. If two models are voting equally for the wrong label the result is also wrong. Thus it quite efficient regarding the time complexity and also in comparison to the already proposed scoring methods like brute-force. An implementation for the major voting system is shown in Listing 4.2

There exists many other possible methods. Every clustering algorithm like k-means would give a relieable result by consider the input labels as cluster points and maps it to natural numbers which then will be observed as emotion. Also rule based classifcation approaches as for example decision trees can be used. Thus this will also exceed the scope of this theses and might be related to future work.

Architecture

All unimodal architectures are already proposed and discussed before in this chapter. Thus regarding to the scoring methods they have to put in the same context especially for the same kind of output to meet the requirements given by the optimization function f. Looking back at the abstract overwievs for each modal as shown in Figure 4.2, Figure 4.4 and Figure 4.8 some tasks are showing up. First of all due to the text-to-text understanding of the natural language models the output has to get mapped back to the one-hot vector which is already given by the audio and video based architectures. Secondly for the neural network approach a dataset out of the results from the original datasets is needed. Therefore every model saves its prediction.

Modal	Architecture	Trainable Parameters
Text	Llama 3	$\approx 4.000.000.000$
	Multi EmoBERT	
	T5	
Audio	Convolution	2.176.474.377
	Audio-Transformer	TBD
Video	Convolution	2.504.047.929
	3D Vision Transformer	1.266.615.309
Multimodal Scoring	Linear Layer	147

Table 4.4.: Overwie of all trainable parameters for each unimodal architecture.

Table 4.4 shows an final overwiev over all trainable parameters for each model and also for the multimodal scoring method featuring a neural network. All models are good comparable regarding their size and amount of trainable weights. The scoring network is the smallest of all thus it includes only a single linear layer which maps three vectors to a single vector.

Chapter 5.

Evaluation

This chapters will show all results for the before described and discussed implementations. Therefore the setup for the evaluation part is shown with a look on accuracy measuring functions especially the precision under recall curve. After this introduction and overwiev the actual results for the experiments are shown, visualized and further discussed.

5.1. Setup

As already explained each model is trained on both datasets for 20 epochs and a 90/10 split in training and testing. Each epoch runs through the whole training data which is randomly shuffled every time for better and more efficient training. The preprocessing follows analogous the implementation presented in the chapter before. The normalizaion is done globally to bring the values in an uniform space between −1 and 1. Every training loop consists of the Adam optimizer and the cross entropy loss function to calculate the gradient. After this each trained model is also tested with each datasat to show if a model trained on CREMA-D is capeable to detect emotion in the MELD data and the other way around. The only data which is not tested is the text data from the CREMA dataset. Due to its lack of a variety of sentences it is not necessary. It consists of only twelve different sentences in around 7000 videos. Every expression is actored in the different emotion. So it is neither learnable nor testable for the further evaluation. After this unimodal detection all predictions for the test and trainings datasets are saved in an external dataset for the multimodal emotion classification so they must not be runned each time for the brute-force, neural network or major vote approach. To measure the accuracy different metrics are used. Initially as example for the text classification the precision is calculated by simply counting how many of the predictions are actually equal to the expected output. Besides it is countet how many and which emotion are wrongly labeled by the model to display a heatmap and to analyize if there are easy mix-ups between similar feature spaces. This is also continued for each test scenario. Thus the calculation of the accuracy changed to a more informative and meaningful metric.

Experimente beschreiben

auf bestimmte Datenarten achten (z.B. kurze Sätze, rauschende Videos oder Audios)

5.1.1. Accuracy Metrics

Precision under recall describes an accuracy metric to evaluate the quality of classification tasks. It provides three scores for the comparability of the values. For the further overview it is necessary to define some variables to later understand the components of this metric.

- $\boldsymbol{T_P}$ $\rightarrow$ True Positive: Amount of all correct classified values.
- $\boldsymbol{T_N}$ $\rightarrow$ True Negative: Amount of all correct unclassified values.
- $\boldsymbol{F_P}$ $\rightarrow$ False Positive: Amount of all falsly predicted labels
- $\boldsymbol{F_N}$ $\rightarrow$ False Negative: Amount of all falsly unpredicted labels

Out of these values the four important metrics can be derived from this. The accuracy shows the proportion of all correct classified labels including the quantity of all true positive and true negative classifcations.

$$\text{Accuracy} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \tag{1}$$

Further while only looking at the true positive values the precision of the model can be calculated. It describes the proportion of all correct classified values.

$$\text{Precision} = \frac{T_P}{T_P + F_P} \tag{2}$$

In addition the recall shows how accurate the model predictet the classes by looking at the true positive and false negative values.

$$\text{Recall} = \frac{T_P}{T_P + F_N} \tag{3}$$

At last to measure the quality of the classifcation model the F1-score can be calculated. It describes the harmonic mean between precision and recall and is denoted by

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

All in all these metrics are normally used for each label instead of for the whole classifier. Thus for a more clear arrangement of the results regarding the amount of different classifier and multimodal combination methods the following results are calculated for all labels in total showing the precision, accuracy, recall and quality of each classifier. Also this global observation of precision under recall helps to ignore imbalances within the datasets. This problem would otherwise occur while evaluating measurements with the MELD dataset.

5.2. Results

Unimodal Results

	CREMA-D				MELD			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Llama 3	-	-	-	-				
Multi-EmoBert	-	-	-	-				
T5	-	-	-	-				

Table 5.1.: Accuracy for unimodal emotion classification in text.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

	CREMA-D				MELD			
	Acc.	Precision	Recall	F1	Acc.	Precision	Recall	F1
Conv.								
Transf.								

(a) Test dataset: CREMA-D

	CREMA-D				MELD			
	Acc.	Precision	Recall	F1	Acc.	Precision	Recall	F1
Conv.								
Transf.								

(b) Test dataset: MELD

Table 5.2.: Accuracy for unimodal emotion classification in audio trained on each dataset.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

	CREMA-D				MELD			
	Acc.	Precision	Recall	F1	Acc.	Precision	Recall	F1
Conv.								
Transf.								

(a) Test dataset: CREMA-D

	CREMA-D				MELD			
	Acc.	Precision	Recall	F1	Acc.	Precision	Recall	F1
Conv.								
Transf.								

(b) Test dataset: MELD

Table 5.3.: Accuracy for unimodal emotion classification in video trained on each dataset.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

Multimodal Results

Chapter 6.

Conclusion

6.1. Summary

6.2. Future Work

Bibliography

[1] W. James, "What is an Emotion?," *Mind*, vol. 9, no. 34, pp. 188–205, 1884, Accessed: Mar. 11, 2024. [Online]. Available: http://www.jstor.org/stable/2246769

[2] R. W. Picard, "Affective computing." MIT Press, Cambridge, Mass. [u.a.], 1997.

[3] K. R. Scherer and P. Ekman, Eds., *Approaches To Emotion*, 1st ed. Psychology Press, 1984.

[4] P. Kumar and M. Vardhan, "PWEBSA: Twitter sentiment analysis by combining Plutchik wheel of emotion and word embedding," *International Journal of Information Technology*, vol. 14, no. 1, pp. 69–77, Feb. 2022.

[5] D. Chafale and A. Pimpalkar, "Review on Developing Corpora for Sentiment Analysis Using Plutchik's Wheel of Emotions with Fuzzy Logic," *International Journal of Computer Sciences and Engineering (IJCSE)*, vol. 2, pp. 14–18, 2014.

[6] M. Müller, *Fundamentals of music processing: Audio, analysis, algorithms, applications*, vol. 5. Springer, 2015.

[7] I. R. Murray and J. L. Arnott, "Applying an analysis of acted vocal emotions to improve the simulation of synthetic speech," *Computer Speech & Language*, vol. 22, no. 2, pp. 107–129, 2008, [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0885230807000393

[8] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "Crema-d: Crowd-sourced emotional multimodal actors dataset," *IEEE transactions on affective computing*, vol. 5, no. 4, pp. 377–390, 2014.

[9] E. Friesen and P. Ekman, "Facial action coding system: a technique for the measurement of facial movement," *Palo Alto*, vol. 3, no. 2, p. 5–6, 1978.

[10] J. C. Hager, P. Ekman, and W. V. Friesen, "Facial action coding system," *Salt Lake City, UT: A Human Face*, p. 8–9, 2002.

[11] B. Wang, A. Wang, F. Chen, Y. Wang, and C.-C. J. Kuo, "Evaluating word embedding models: methods and experimental results," *APSIPA Transactions on Signal and Information Processing*, vol. 8, p. e19, 2019, doi: 10.1017/ATSIP.2019.12.

[12] D. Karani, "Introduction to word embedding and Word2Vec," *Towards Data Science*, vol. 1, 2018.

[13] Y. Li and T. Yang, "Word Embedding for Understanding Natural Language: A Survey," in *Guide to Big Data Applications*, S. Srinivasan, Ed., Cham: Springer International Publishing, 2018, pp. 83–104.

[14] K. Choudhary and R. Beniwal, "Xplore Word Embedding Using CBOW Model and Skip-Gram Model," in *2021 7th International Conference on Signal Processing and Communication (ICSC)*, 2021, pp. 267–270. doi: 10.1109/ICSC53193.2021.9673321.

[15] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," *arXiv preprint arXiv:1301.3781*, 2013.

[16] X. Rong, "word2vec Parameter Learning Explained." 2016.

[17] L. Ma and Y. Zhang, "Using Word2Vec to process big text data," in *2015 IEEE International Conference on Big Data (Big Data)*, 2015, pp. 2895–2897. doi: 10.1109/BigData.2015.7364114.

[18] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." 2019.

[19] E. Fredkin, "Trie memory," *Commun. ACM*, vol. 3, no. 9, pp. 490–499, Sep. 1960, doi: 10.1145/367390.367400.

[20] X. Song, A. Salcianu, Y. Song, D. Dopson, and D. Zhou, "Fast WordPiece Tokenization," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds., Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 2089–2103. doi: 10.18653/v1/2021.emnlp-main.160.

[21] C. Fellbaum and K. Brown, "Encyclopedia of language and linguistics." Elsevier Oxford, 2005.

[22] B. Liu, "Text sentiment analysis based on CBOW model and deep learning in big data environment," *Journal of ambient intelligence and humanized computing*, vol. 11, no. 2, pp. 451–458, 2020.

[23] E. S. Lubana, R. Dick, and H. Tanaka, "Beyond BatchNorm: Towards a Unified Understanding of Normalization in Deep Learning," in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., Curran Associates, Inc., 2021, pp. 4778–4791. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2021/file/2578eb9cdf020730f77793e8b58e165a-Paper.pdf

[24] L. Medsker and L. C. Jain, *Recurrent neural networks: design and applications*. CRC press, 1999.

[25] A. Graves, "Long Short-Term Memory," in *Supervised Sequence Labelling with Recurrent Neural Networks*, Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 37–45. doi: 10.1007/978-3-642-24797-2_4.

[26] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling." 2014.

[27] A. Vaswani *et al.*, "Attention Is All You Need." 2023.

[28] OpenAI *et al.*, "GPT-4 Technical Report." 2024.

[29] S. L. Smith, A. Brock, L. Berrada, and S. De, "ConvNets Match Vision Transformers at Scale." 2023.

[30] R. Sutton, "The bitter lesson, 2019," *URL http://www. incompleteideas. net/IncIdeas/BitterLesson. html*, 2019.

[31] J. Kaplan *et al.*, "Scaling Laws for Neural Language Models." 2020.

[32] S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, and R. Harshman, "Indexing by latent semantic analysis," *Journal of the American Society for Information Science*, vol. 41, no. 6, pp. 391–407, 1990, doi: https://doi.org/10.1002/(SICI)1097-4571(199009)41:6<391::AID-ASI1>3.0.CO;2-9.

[33] V. Klema and A. Laub, "The singular value decomposition: Its computation and some applications," *IEEE Transactions on Automatic Control*, vol. 25, no. 2, pp. 164–176, 1980, doi: 10.1109/TAC.1980.1102314.

[34] A. Agrawal and A. An, "Unsupervised emotion detection from text using semantic and syntactic relations," in *2012 IEEE/WIC/ACM International Conferences on Web Intelligence and Intelligent Agent Technology*, 2012, pp. 346–353.

[35] S. D. Gollapalli, P. Rozenshtein, and S. K. Ng, "ESTeR: Combining word co-occurrences and word associations for unsupervised emotion detection," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 1043–1056.

[36] Y.-c. Park, J.-G. Jang, and U. Kang, "Fast Partial Fourier Transform." 2020.

[37] L. Durak and O. Arikan, "Short-time Fourier transform: two fundamental properties and an optimal implementation," *IEEE Transactions on Signal Processing*, vol. 51, no. 5, pp. 1231–1242, 2003.

[38] L. Wang *et al.*, "Towards Learning Universal Audio Representations." 2022.

[39] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, "PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition." 2020.

[40] Y. M. Costa, L. S. Oliveira, and C. N. Silla, "An evaluation of Convolutional Neural Networks for music classification using spectrograms," *Applied Soft Computing*, vol. 52, pp. 28–38, 2017, doi: https://doi.org/10.1016/j.asoc.2016.12.024.

[41] M. French and R. Handy, "Spectrograms: Turning Signals into Pictures," *Journal of Engineering Technology*, vol. 24, pp. 32–35, 2007.

[42] P. Verma and J. Berger, "Audio Transformers:Transformer Architectures For Large Scale Audio Understanding. Adieu Convolutions." 2021.

[43] J. Wülfing and M. A. Riedmiller, "Unsupervised Learning of Local Features for Music Classification.," in *ISMIR*, 2012, pp. 139–144.

[44] Y.-C. Lu, C.-W. Wu, C.-T. Lu, and A. Lerch, "An Unsupervised Approach to Anomaly Detection in Music Datasets," in *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval*, in SIGIR '16. Pisa, Italy: Association for Computing Machinery, 2016, pp. 749–752. doi: 10.1145/2911451.2914700.

[45] M. Jogin, Mohana, M. S. Madhulika, G. D. Divya, R. K. Meghana, and S. Apoorva, "Feature Extraction using Convolution Neural Networks (CNN) and Deep Learning," in *2018 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, 2018, pp. 2319–2323. doi: 10.1109/RTEICT42901.2018.9012507.

[46] J. Selva, A. S. Johansen, S. Escalera, K. Nasrollahi, T. B. Moeslund, and A. Clapés, "Video Transformers: A Survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 11, pp. 12922–12943, 2023, doi: 10.1109/TPAMI.2023.3243465.

[47] F. Krause, *Video Emotion Classification utilizing Temporal Difference Images*. Master Thesis, Otto-von-Guericke University Magdeburg, 2023.

[48] A. Kumar, A. Kumar, V. Vineet, and Y. Rawat, "Benchmarking self-supervised video representation learning," p. , 2023.

[49] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lučić, and C. Schmid, "ViViT: A Video Vision Transformer." 2021.

[50] H. Akbari *et al.*, "VATT: Transformers for Multimodal Self-Supervised Learning from Raw Video, Audio and Text," *ArXiv*, 2021, [Online]. Available: https://api.semanticscholar.org/CorpusID:233346984

[51] G. Patil and P. Suja, "Emotion recognition from 3D videos using optical flow method," in *2017 International Conference On Smart Technologies For Smart Nation (SmartTechCon)*, 2017, pp. 825–829. doi: 10.1109/SmartTechCon.2017.8358488.

[52] J. Haddad, O. Lezoray, and P. Hamel, "3D-CNN for Facial Emotion Recognition in Videos," in *Advances in Visual Computing*, G. Bebis, Z. Yin, E. Kim, J. Bender, K. Subr, B. C. Kwon, J. Zhao, D. Kalkofen, and G. Baciu, Eds., Cham: Springer International Publishing, 2020, pp. 298–309.

[53] L. Zhang, Y. Tong, and Q. Ji, "Active Image Labeling and Its Application to Facial Action Labeling," in *Computer Vision – ECCV 2008*, D. Forsyth, P. Torr, and A. Zisserman, Eds., Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 706–719.

[54] Y. Tong, W. Liao, and Q. Ji, "Facial Action Unit Recognition by Exploiting Their Dynamic and Semantic Relationships," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 29, no. 10, pp. 1683–1699, 2007, doi: 10.1109/TPAMI.2007.1094.

[55] B. Martinez, M. F. Valstar, B. Jiang, and M. Pantic, "Automatic Analysis of Facial Actions: A Survey," *IEEE Transactions on Affective Computing*, vol. 10, no. 3, pp. 325–347, 2019, doi: 10.1109/TAFFC.2017.2731763.

[56] M. Valstar, M. Pantic, and I. Patras, "Motion history for facial action detection in video," in *2004 IEEE International Conference on Systems, Man and Cybernetics (IEEE Cat. No.04CH37583)*, 2004, pp. 635–640. doi: 10.1109/ICSMC.2004.1398371.

[57] H. Shimodaira, K.-i. Noma, M. Nakai, and S. Sagayama, "Dynamic Time-Alignment Kernel in Support Vector Machine," in *Advances in Neural Information Processing Systems*, T. Dietterich, S. Becker, and Z. Ghahramani, Eds., MIT Press, 2001, p. . [On-

line]. Available: https://proceedings.neurips.cc/paper_files/paper/2001/file/a869ccbcbd9568808b8497e28275c7c8-Paper.pdf

[58] F. Zhou, F. De la Torre, and J. F. Cohn, "Unsupervised discovery of facial events," in *2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2010, pp. 2574–2581. doi: 10.1109/CVPR.2010.5539966.

[59] T. Warren Liao, "Clustering of time series data—a survey," *Pattern Recognition*, vol. 38, no. 11, pp. 1857–1874, 2005, doi: https://doi.org/10.1016/j.patcog.2005.01.025.

[60] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "Meld: A multimodal multi-party dataset for emotion recognition in conversations," *arXiv preprint arXiv:1810.02508*, 2018.

[61] C. Busso *et al.*, "IEMOCAP: Interactive emotional dyadic motion capture database," *Language resources and evaluation*, vol. 42, pp. 335–359, 2008.

[62] V. Vielzeuf, S. Pateux, and F. Jurie, "Temporal multimodal fusion for video emotion classification in the wild," in *Proceedings of the 19th ACM International Conference on Multimodal Interaction*, in ICMI '17. Glasgow, UK: Association for Computing Machinery, 2017, pp. 569–576. doi: 10.1145/3136755.3143011.

[63] A. Dhall, R. Goecke, S. Lucey, and T. Gedeon, "Collecting Large, Richly Annotated Facial-Expression Databases from Movies," *IEEE MultiMedia*, vol. 19, no. 3, pp. 34–41, 2012, doi: 10.1109/MMUL.2012.26.

[64] A. S. Patwardhan, "Multimodal mixed emotion detection," in *2017 2nd International Conference on Communication and Electronics Systems (ICCES)*, 2017, pp. 139–143. doi: 10.1109/CESYS.2017.8321250.

[65] M. G. Huddar, S. S. Sannakki, and V. S. Rajpurohit, "Multi-level feature optimization and multimodal contextual fusion for sentiment analysis and emotion classification," *Computational Intelligence*, vol. 36, no. 2, pp. 861–881, 2020, doi: https://doi.org/10.1111/coin.12274.

[66] C. Yang, F. Chen, and S. Tian, "Grouped Genetic Algorithm Based Optimal Tests Selection for System with Multiple Operation Modes," *Journal of Electronic Testing*, vol. 33, p. , 2017, doi: 10.1007/s10836-017-5672-y.

[67] N. Majumder, D. Hazarika, A. Gelbukh, E. Cambria, and S. Poria, "Multimodal sentiment analysis using hierarchical fusion with context modeling," *Knowledge-Based Systems*, vol. 161, pp. 124–133, 2018, doi: https://doi.org/10.1016/j.knosys.2018.07.041.

[68] H. Pham, T. Manzini, P. P. Liang, and B. Poczós, "Seq2Seq2Sentiment: Multimodal Sequence to Sequence Models for Sentiment Analysis," in *Proceedings of Grand Challenge and Workshop on Human Multimodal Language (Challenge-HML)*, A. Zadeh, P. P. Liang, L.-P. Morency, S. Poria, E. Cambria, and S. Scherer, Eds., Melbourne, Australia: Association for Computational Linguistics, Jul. 2018, pp. 53–63. doi: 10.18653/v1/W18-3308.

[69] J. Pennington, R. Socher, and C. D. Manning, "Glove: Global vectors for word representation," in *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, 2014, pp. 1532–1543.

[70] G. Degottex, J. Kane, T. Drugman, T. Raitio, and S. Scherer, "COVAREP — A collaborative voice analysis repository for speech technologies," in *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2014, pp. 960–964. doi: 10.1109/ICASSP.2014.6853739.

[71] A. Zadeh, M. Chen, S. Poria, E. Cambria, and L.-P. Morency, "Tensor Fusion Network for Multimodal Sentiment Analysis." 2017.

[72] G. Xu, W. Li, and J. Liu, "A social emotion classification approach using multi-model fusion," *Future Generation Computer Systems*, vol. 102, pp. 347–356, 2020, doi: https://doi.org/10.1016/j.future.2019.07.007.

[73] S. Sonnenburg, G. Rätsch, C. Schäfer, and B. Schölkopf, "Large scale multiple kernel learning," *The Journal of Machine Learning Research*, vol. 7, pp. 1531–1565, 2006.

[74] "Introducing Meta Llama 3: The most capable openly available LLM to date. ai.meta.com." 2023.

[75] J. Li and L. Xiao, "Multi-emotion Recognition Using Multi-EmoBERT and Emotion Analysis in Fake News," in *Proceedings of the 15th ACM Web Science Conference 2023*, 2023, pp. 128–135.

[76] C. Raffel *et al.*, "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer." 2023.

[77] B. Hammarberg, B. Fritzell, J. Gaufin, J. Sundberg, and L. Wedin, "Perceptual and acoustic correlates of abnormal voice qualities," *Acta oto-laryngologica*, vol. 90, no. 1–6, pp. 441–451, 1980.

[78] M. Berg, M. Fuchs, K. Wirkner, M. Loeffler, C. Engel, and T. Berger, "The Speaking Voice in the General Population: Normative Data and Associations to Sociodemographic and Lifestyle Factors," *Journal of Voice*, vol. 31, no. 2, p. 257–258, 2017, doi: https://doi.org/10.1016/j.jvoice.2016.06.001.

[79] A. Dosovitskiy *et al.*, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." [Online]. Available: https://arxiv.org/abs/2010.11929

[80] L. Beyer, X. Zhai, and A. Kolesnikov, "Better plain ViT baselines for ImageNet-1k." [Online]. Available: https://arxiv.org/abs/2205.01580

Statement of Authorship

I herewith assure that I wrote the present thesis independently, that the thesis has not been partially or fully submitted as graded academic work and that I have used no other means than the ones indicated. I have indicated all parts of the work in which sources are used according to their wording or to their meaning.

I am aware of the fact that violations of copyright can lead to injunctive relief and claims for damages of the author as well as a penalty by the law enforcement agency.

Magdeburg, September 13, 2024

_____

Signature